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(54) **ELECTRONIC ASSEMBLY HAVING A
CABLE CONNECTOR MODULE**

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(71) Applicant: **TE Connectivity Solutions GmbH**,
Schaffhausen (CH)

(72) Inventors: **Chad William Morgan**, Carneys Point,
NJ (US); **Bruce Allen Champion**,
Camp Hill, PA (US); **John Joseph**
Consoli, Harrisburg, PA (US)

(73) Assignee: **TE CONNECTIVITY SOLUTIONS**
GmbH, Schaffhausen (CH)

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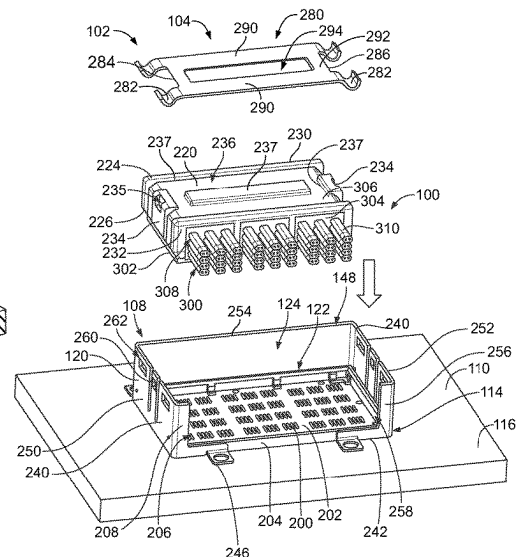
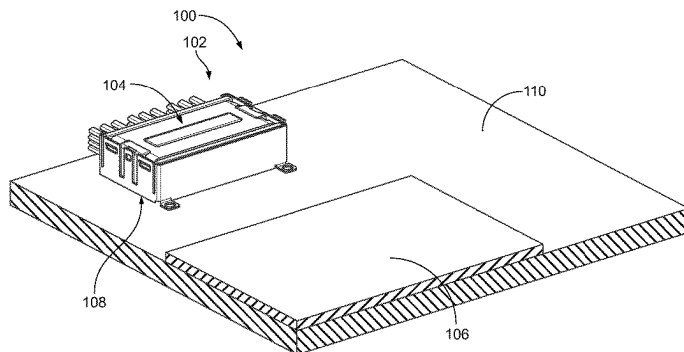
Primary Examiner — Abdullah A Riyami

Assistant Examiner — Nelson R. Burgos-Guntin

(57) **ABSTRACT**

An electronic assembly includes a socket assembly including a cage mounted to a circuit board and a socket connector received in a cavity of the cage. The socket connector includes a substrate holding interposer contacts with compressible upper and lower contact portions. The electronic assembly includes a cable connector module received in the cavity of the cage and coupled to the socket connector. The cable connector module includes a housing coupled to the cage, a circuit card received in the cavity of the housing, and cable assemblies coupled to the circuit card. The circuit card is coupled to the socket connector.

21 Claims, 11 Drawing Sheets



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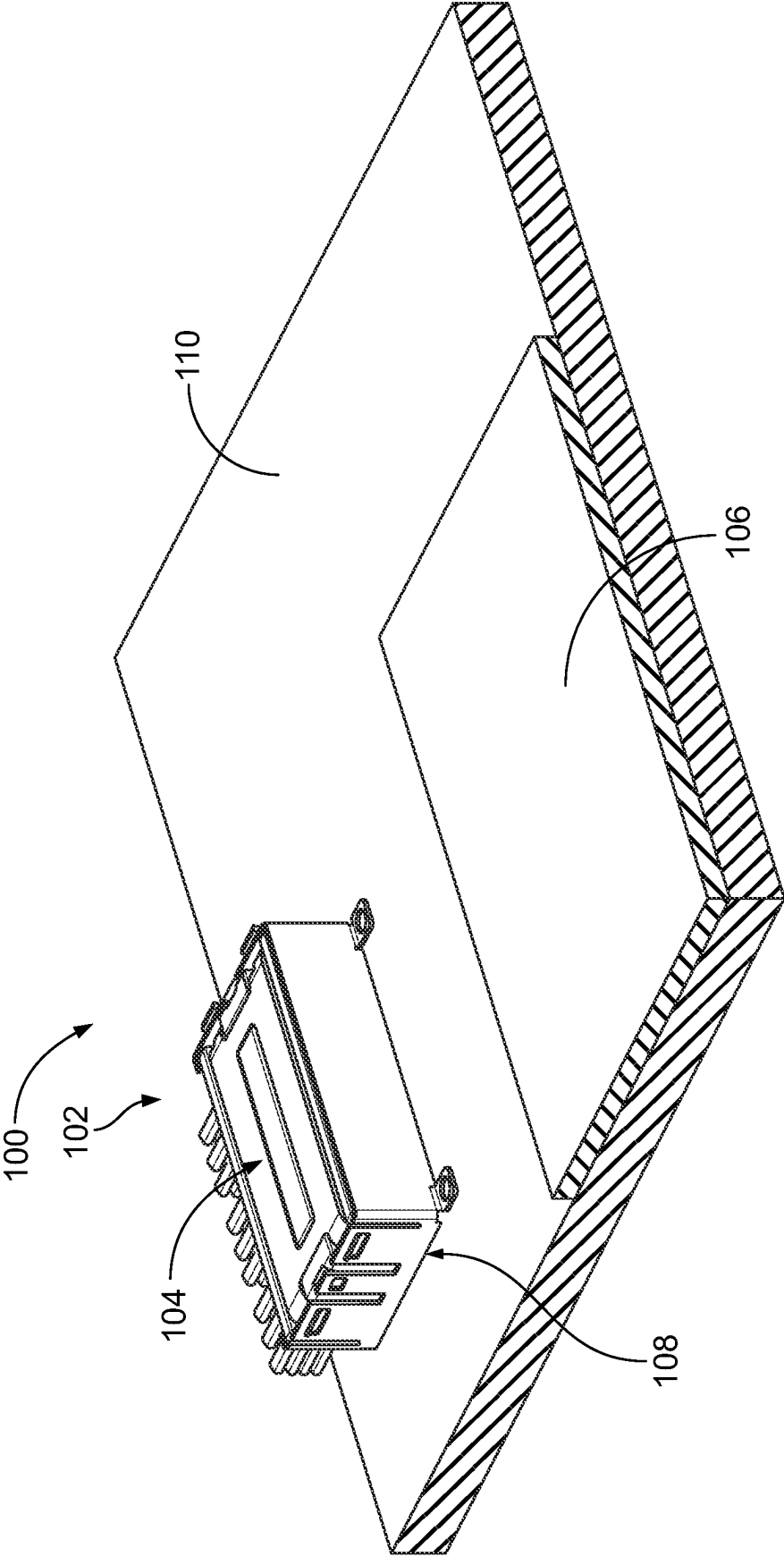


FIG. 1

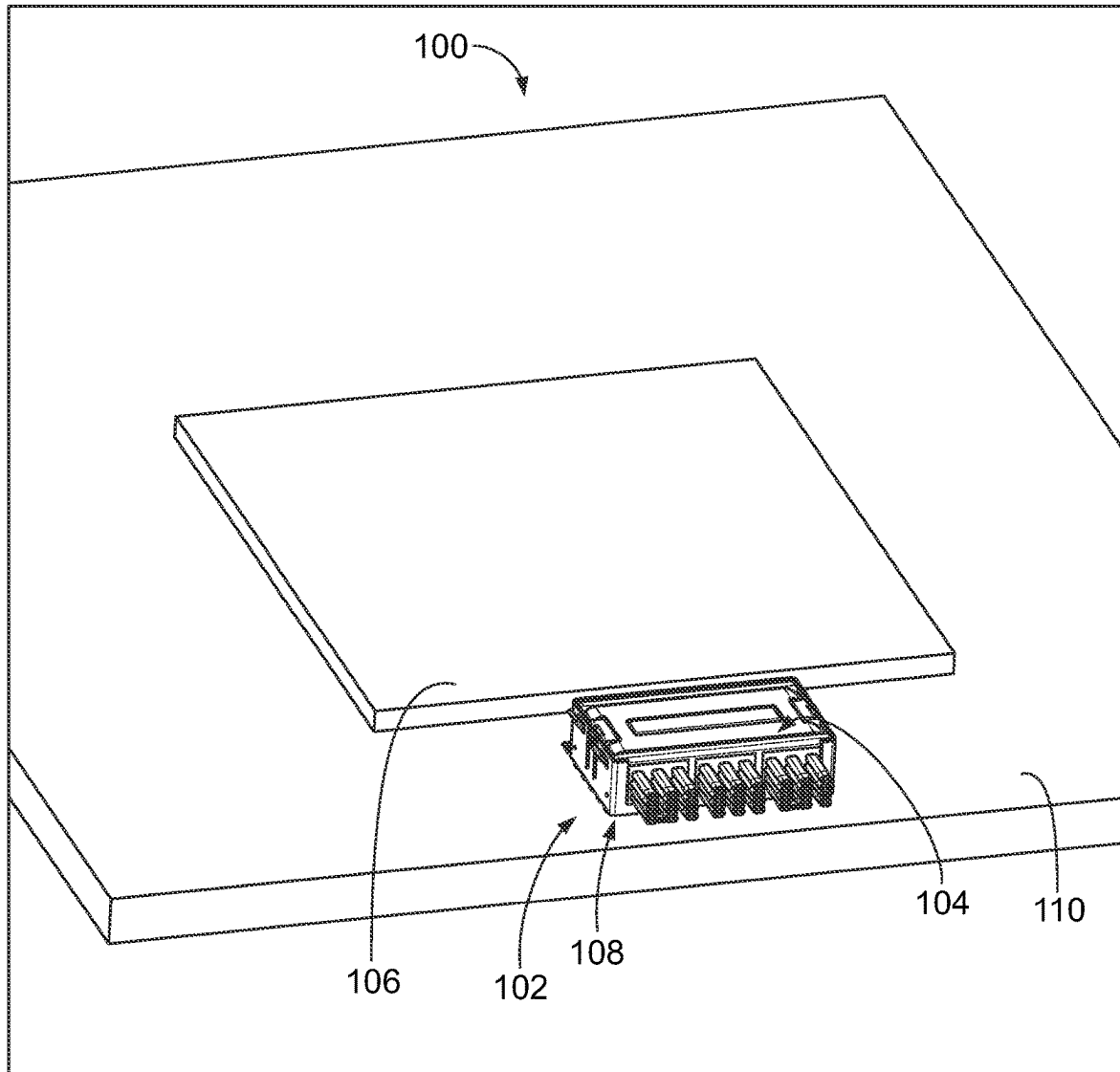


FIG. 2

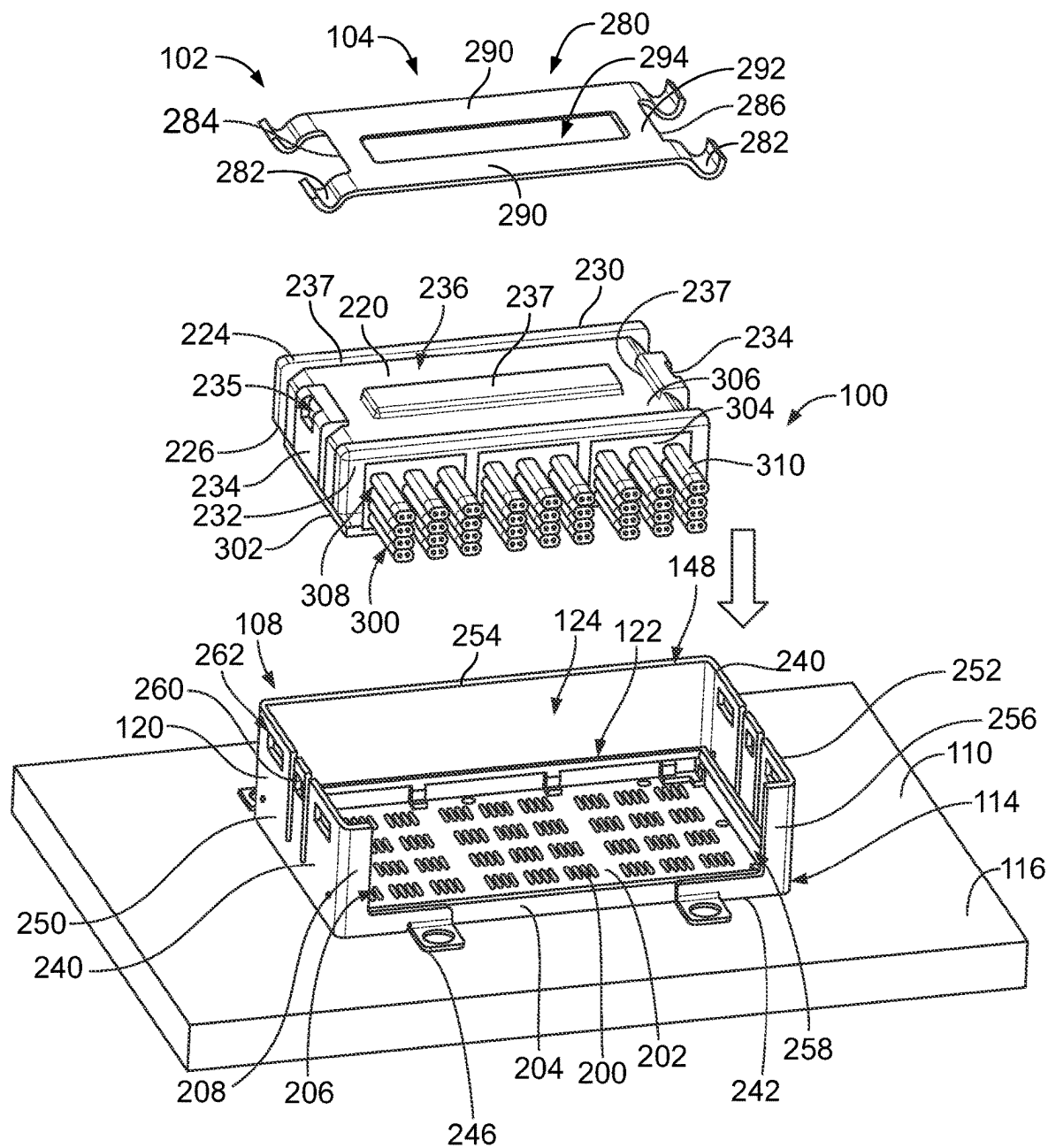


FIG. 3

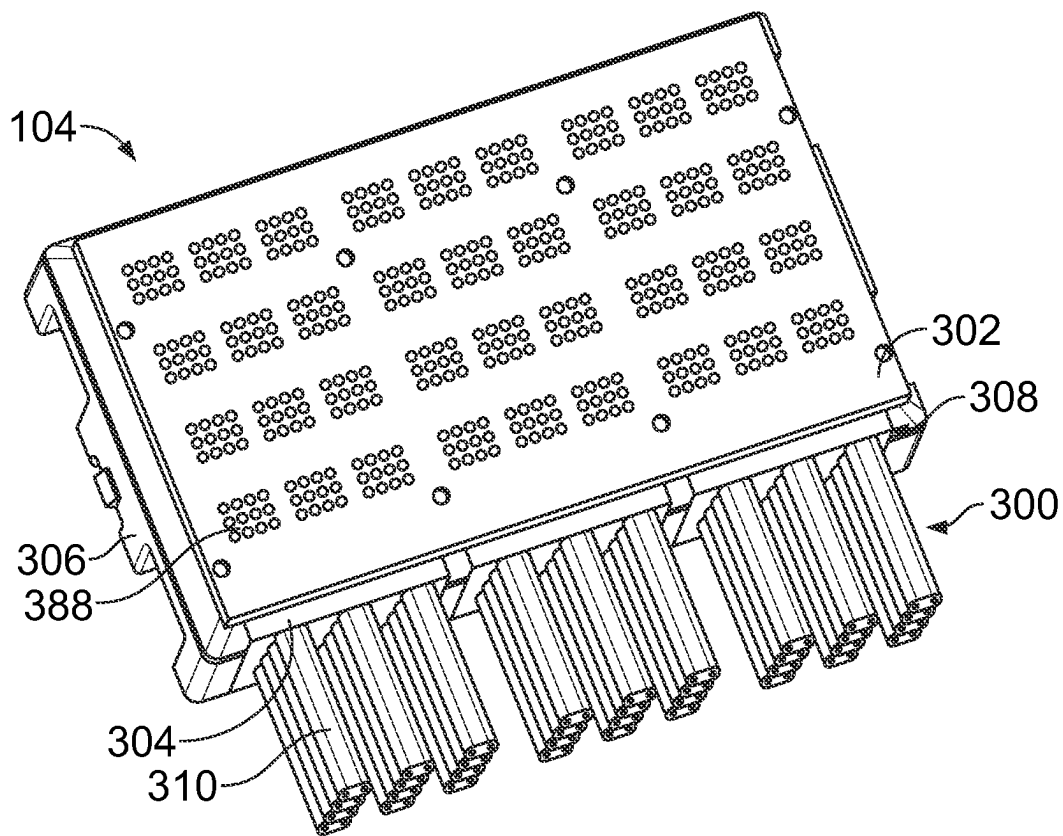


FIG. 4

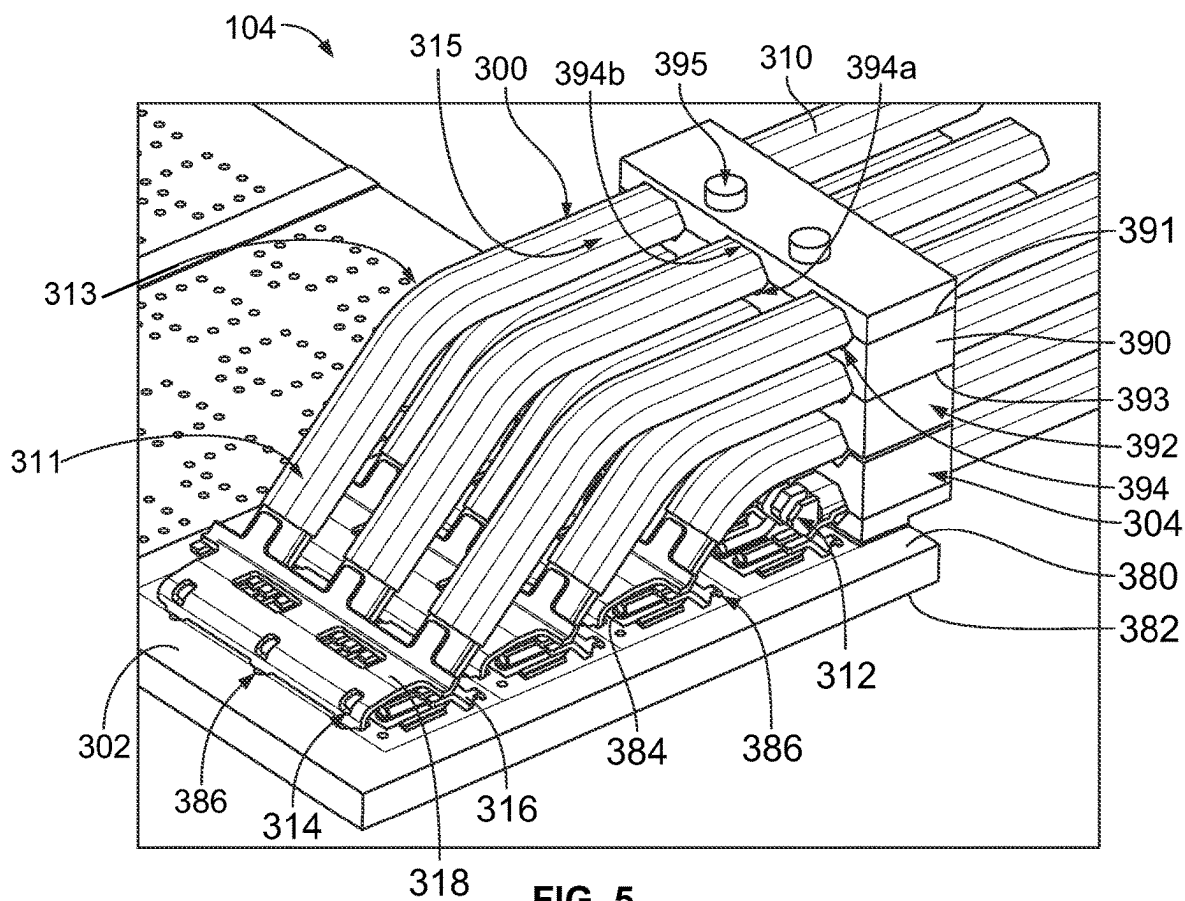


FIG. 5

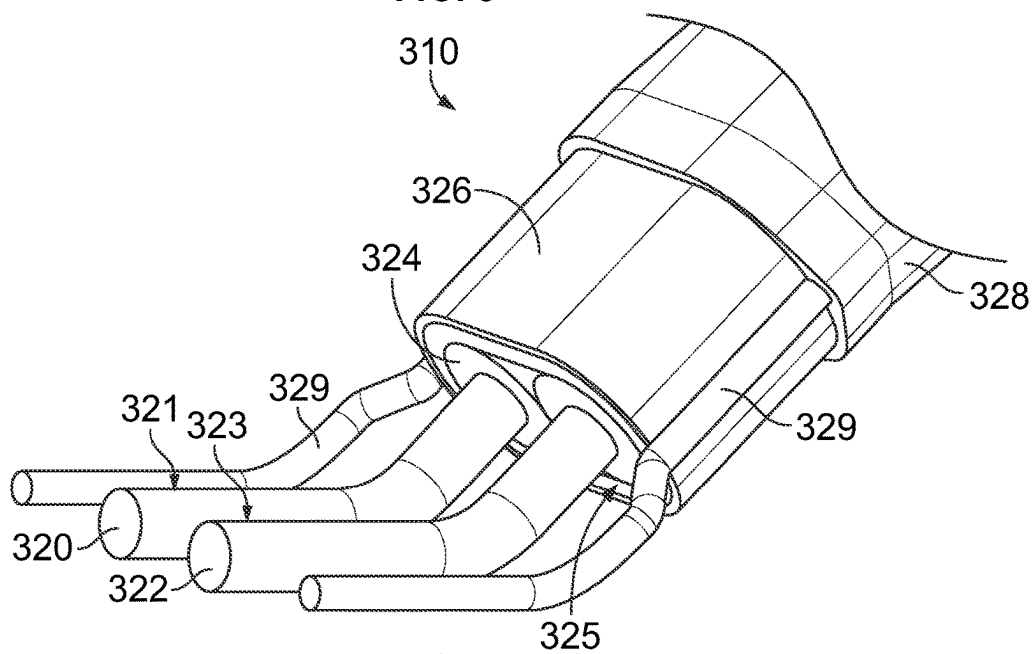


FIG. 6

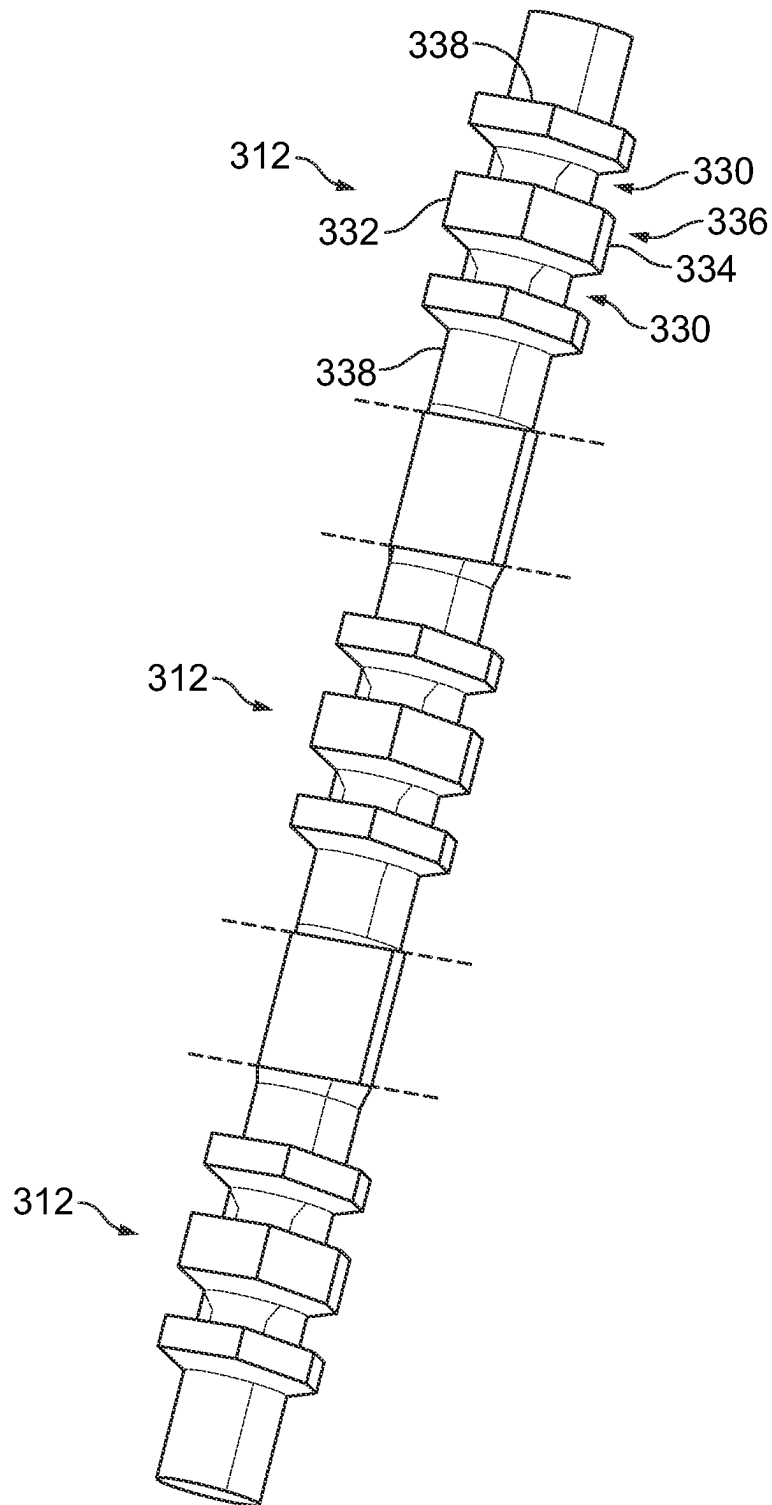


FIG. 7

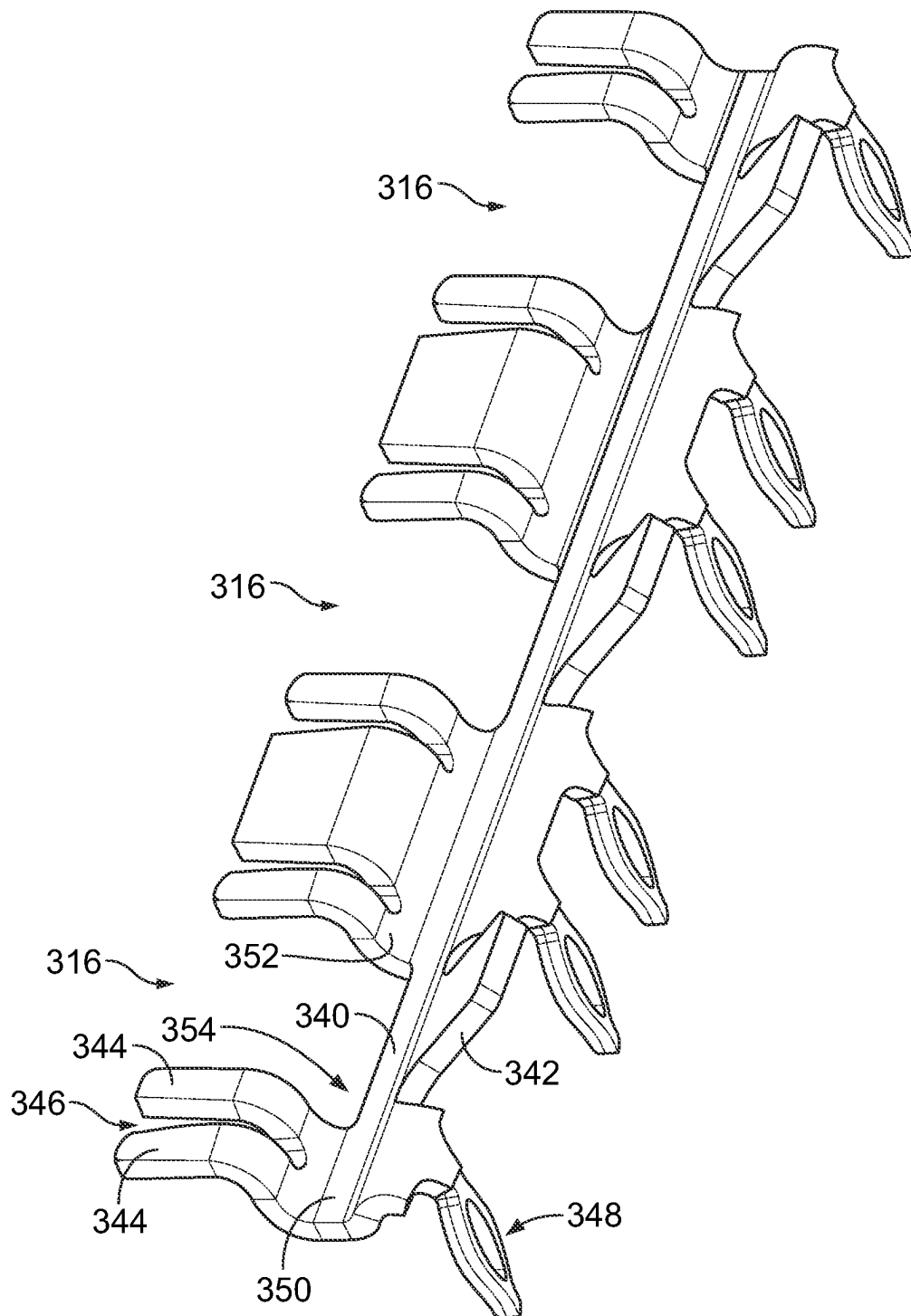


FIG. 8

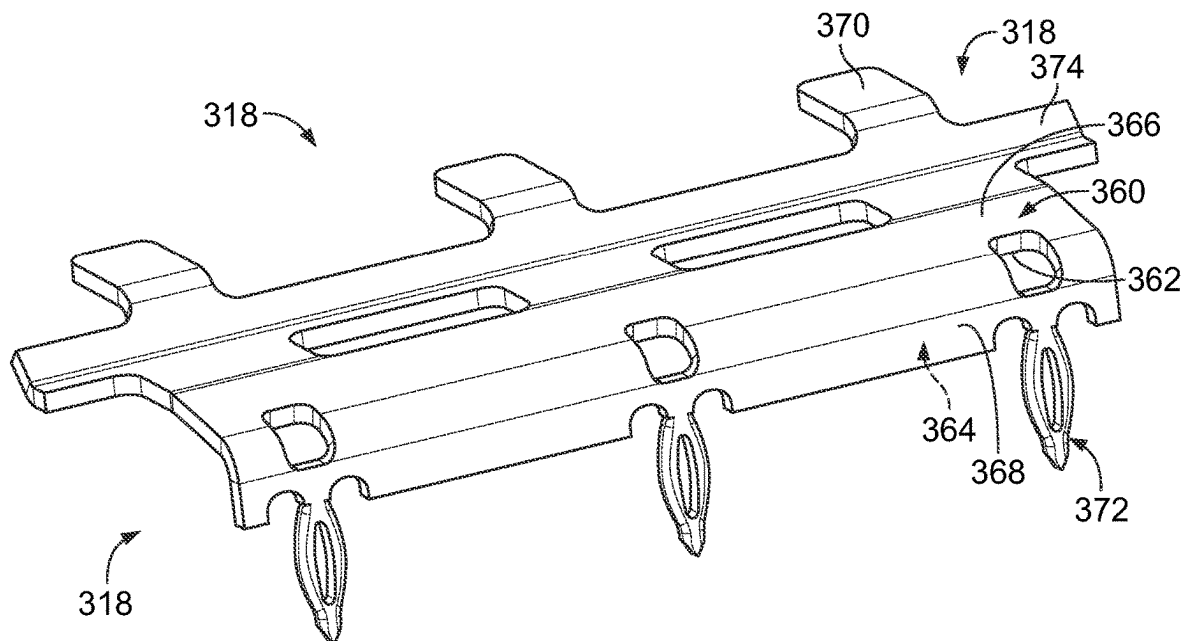


FIG. 9

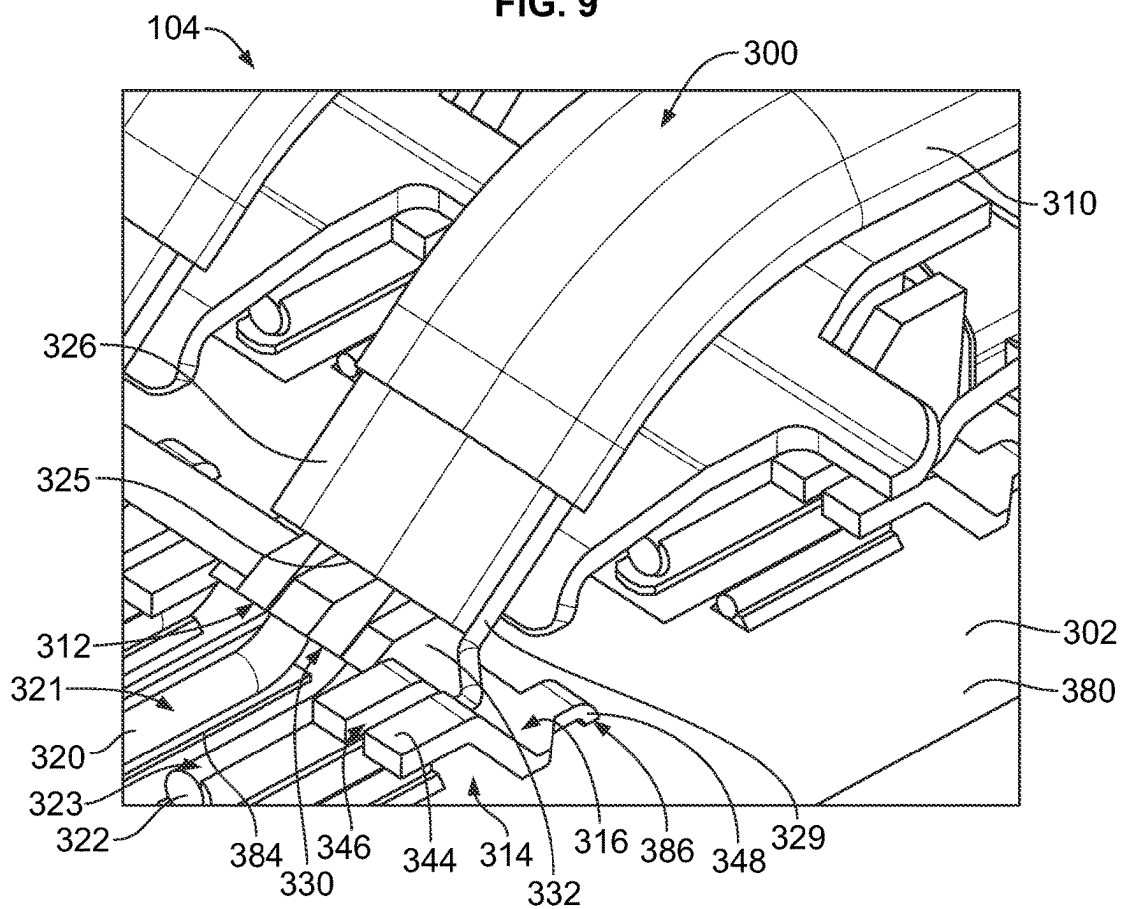


FIG. 10

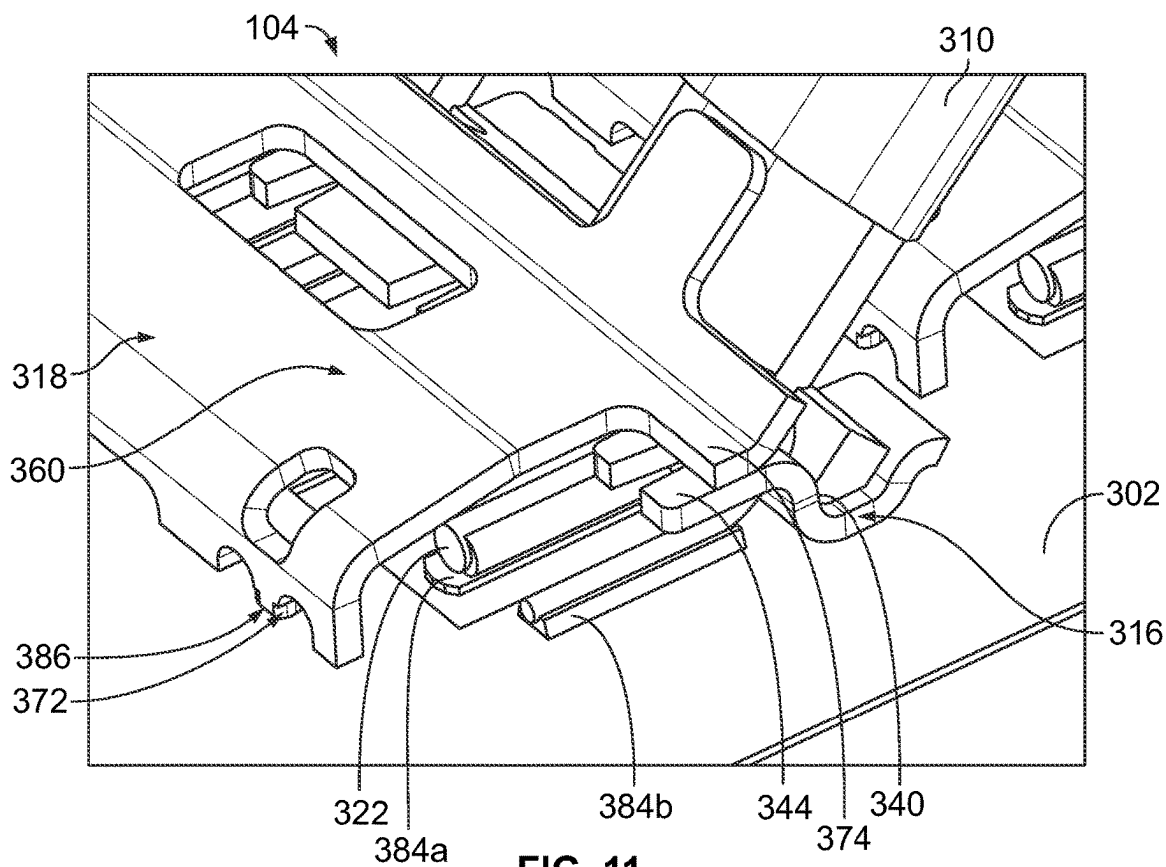


FIG. 11

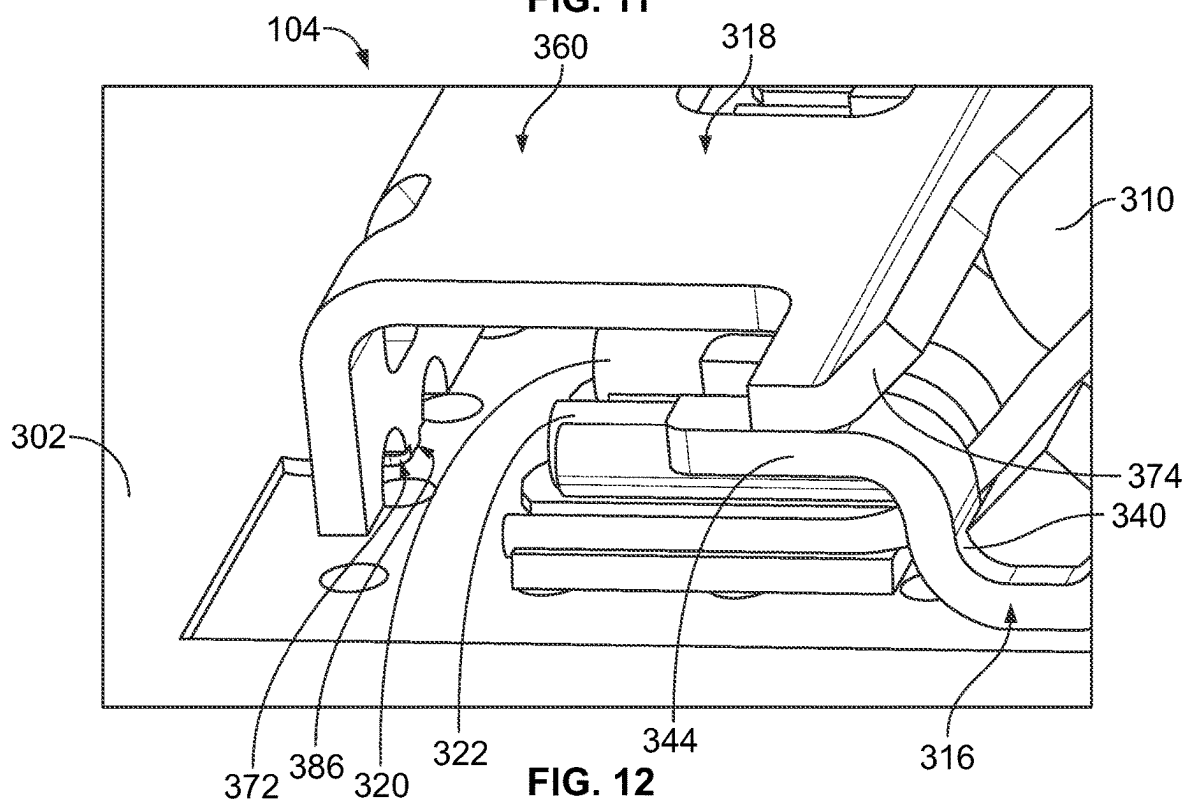
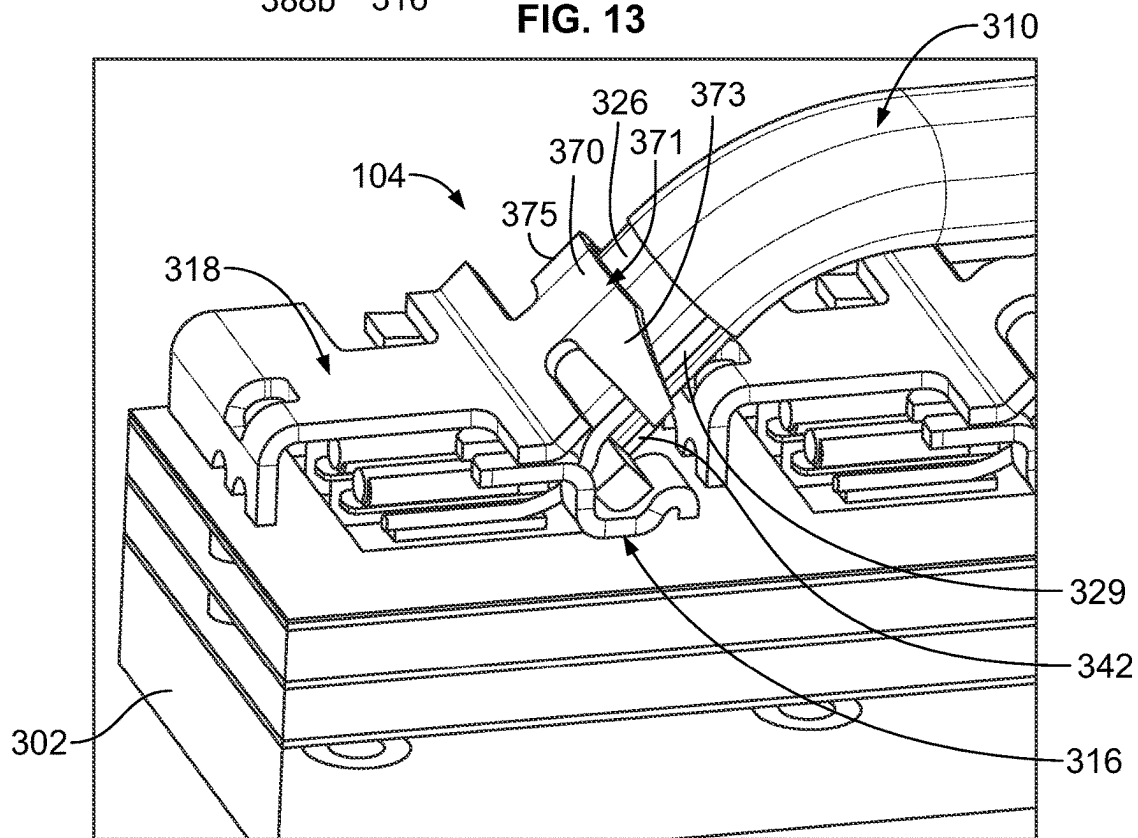
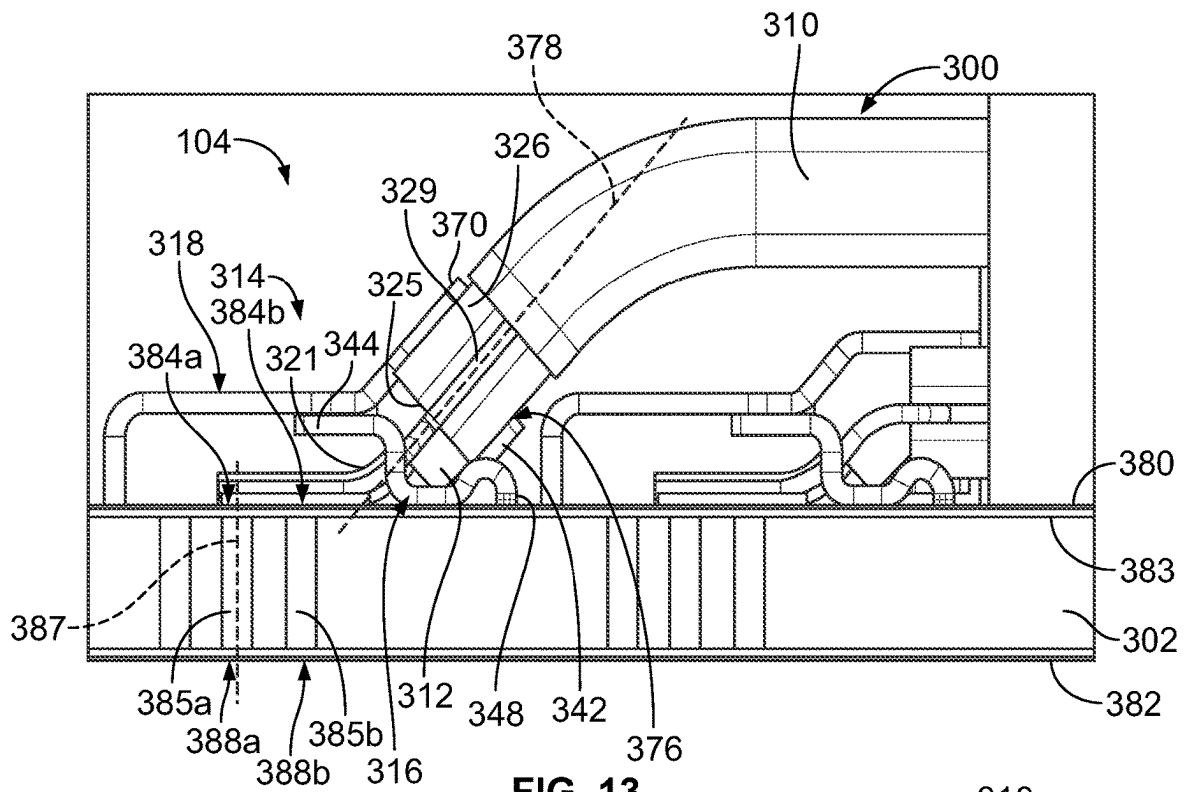


FIG. 12



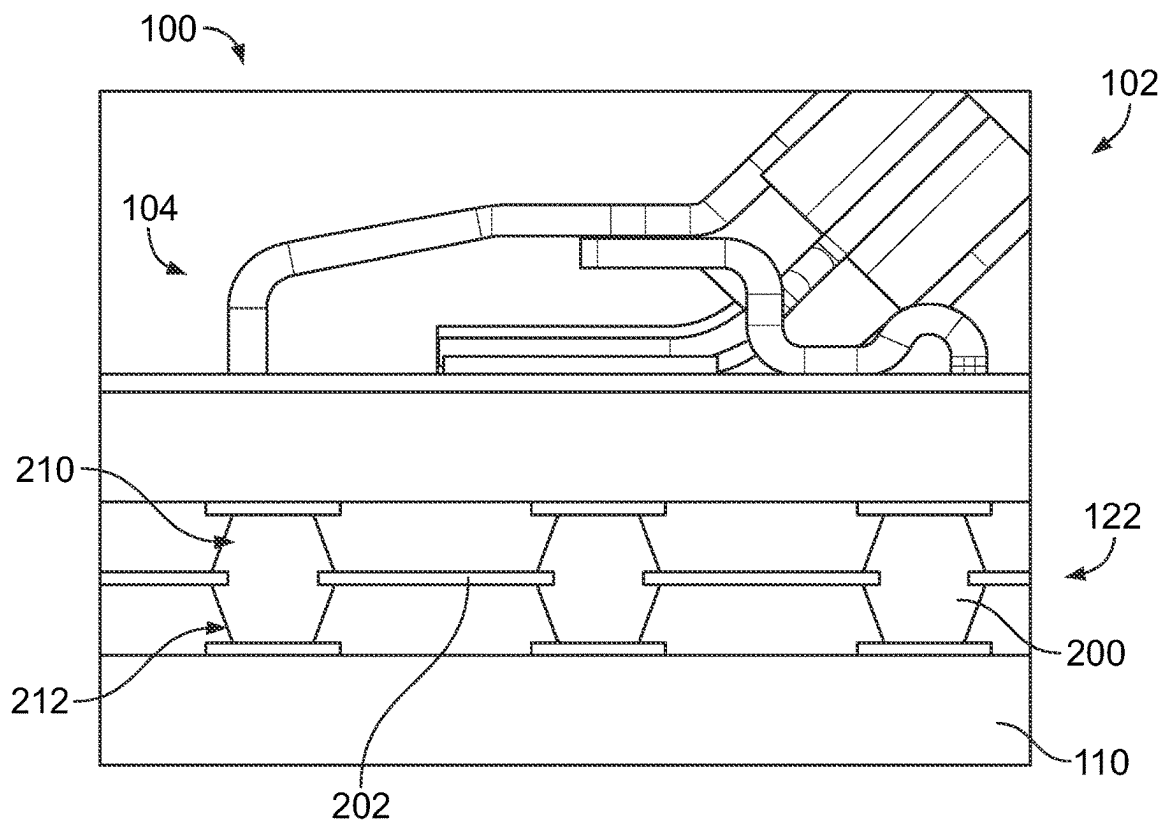


FIG. 15

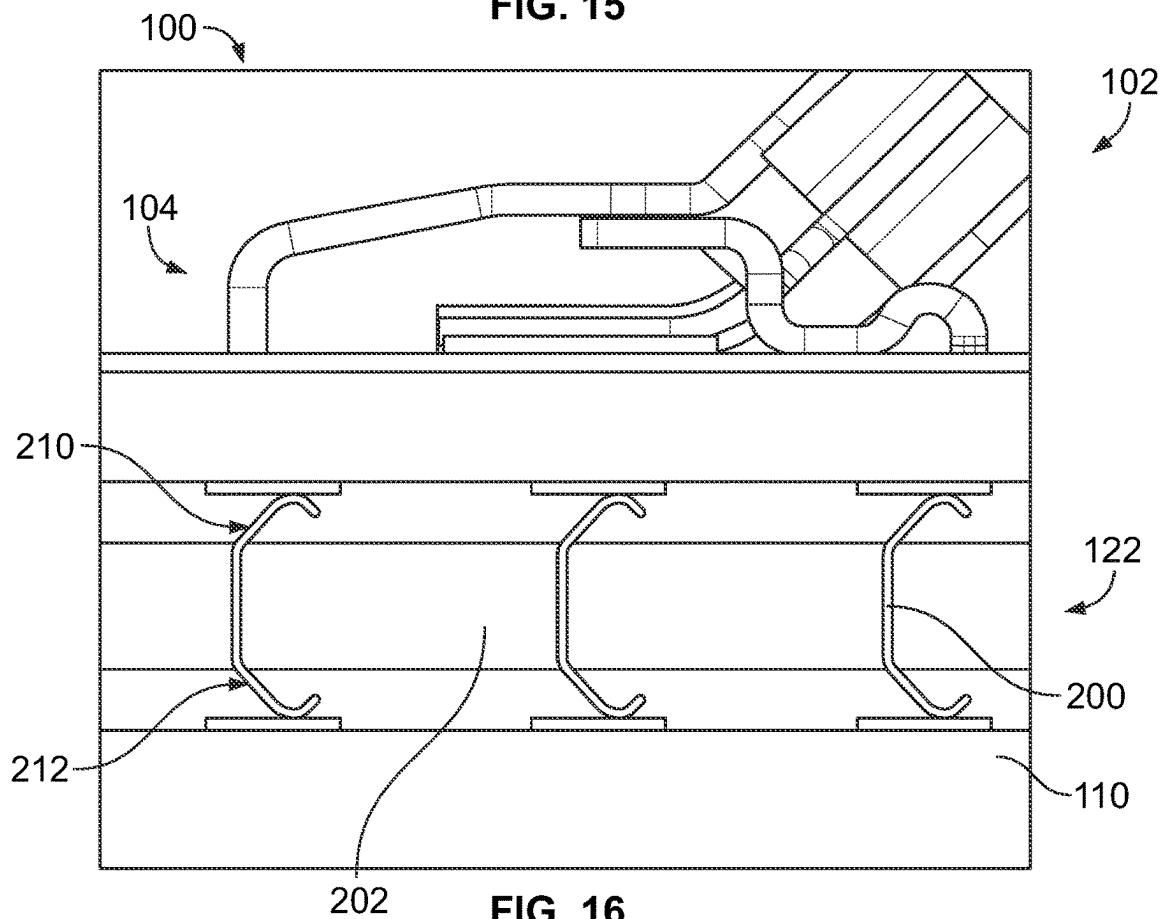


FIG. 16

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ELECTRONIC ASSEMBLY HAVING A CABLE CONNECTOR MODULE

BACKGROUND OF THE INVENTION

The subject matter herein relates generally to communication systems.

There is an ongoing trend toward smaller, lighter, and higher performance communication components and higher density systems, such as for ethernet switches or other system components. Typically, the system includes an electronic package coupled to a circuit board, such as through a socket connector. Electrical signals are routed between the electronic package and the circuit board. The electrical signals are then routed along traces on the circuit board to another component, such as a transceiver connector. The long electrical paths through the host circuit board reduce electrical performance of the system. Additionally, losses are experienced between the connector interfaces and along the electrical signal paths of the transceivers. Conventional systems are struggling with meeting signal and power output from the electronic package. Some known systems utilize an electronic assembly having cable assemblies to transmit the signals along cables rather than signal traces along the host circuit board. However, the electronic assembly includes numerous cables terminated to a circuit card. There is a need to increase the density of the cables and the contact pads on the circuit card to reduce the overall size of the electronic assembly. However, there are limits to spacing of the contact pads to allow routing of the cables from the circuit card with conventional cable termination techniques. For example, ample spacing is needed between rows of the circuit cards to allow routing of the cables along the circuit card. Additionally, as data speeds increase, the grounding structure at the interface between the cables and the circuit card is proving ineffective, particularly at higher frequencies.

BRIEF DESCRIPTION OF THE INVENTION

In one embodiment, an electronic assembly is provided and includes a socket assembly including a cage configured to be mounted to a circuit board. The cage having a cavity. The socket assembly includes a socket connector received in the cavity of the cage. The socket connector includes a substrate holding interposer contacts. Each interposer contact includes an upper contact portion and a lower contact portion. The upper contact portion is compressible. The lower contact portion is compressible. The lower contact portion is terminated to a board contact of the circuit board. The electronic assembly includes a cable connector module received in the cavity of the cage and coupled to the socket connector. The cable connector module includes a housing having a cavity. The housing is coupled to the cage. The cable connector module includes a circuit card held by the housing. The circuit card has signal contact pads at an upper surface of the circuit card and mating contact pads at a lower surface of the circuit card. The mating contact pads are coupled to the upper contact portions of corresponding interposer contacts. The signal contact pads are arranged in pairs in a plurality of rows. The cable connector module includes cable assemblies coupled to the circuit card.

In another embodiment, an electronic assembly is provided and includes a socket assembly including a cage configured to be mounted to a circuit board. The cage has walls extending between a top and a bottom of the cage. The walls define a cavity. The socket assembly includes a socket connector received in the cavity at the bottom of the cage.

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The socket connector is configured to be mounted to the circuit board. The socket connector includes a substrate holding interposer contacts. Each interposer contact includes an upper contact portion and a lower contact portion. The lower contact portion is terminated to a board contact of the circuit board. The electronic assembly includes a cable connector module received in the cavity of the cage and coupled to the socket connector. The cable connector module includes a housing having a cavity. The cable connector module includes a circuit card held by the housing. The circuit card having signal contact pads at an upper surface of the circuit card and mating contact pads at a lower surface of the circuit card. The mating contact pads are coupled to the upper contact portions of corresponding interposer contacts. The signal contact pads are arranged in pairs in a plurality of rows. The cable connector module includes cable assemblies coupled to the circuit card. The electronic assembly includes a spring clip coupled to the walls of the cage. The spring clip engages the cable connector module to hold the cable connector module in the cavity of the cage. The spring clip presses the cable connector module into the socket connector to mate the mating contact pads with the upper contact portions of the interposer contacts.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a front perspective view of a communication system having an electronic assembly in accordance with an exemplary embodiment.

FIG. 2 is a rear perspective view of a communication system having an electronic assembly in accordance with an exemplary embodiment.

FIG. 3 is an exploded view of the communication system in accordance with an exemplary embodiment showing the electronic assembly and the circuit board.

FIG. 4 is a bottom perspective view of the cable connector module 104 in accordance with an exemplary embodiment.

FIG. 5 is a top perspective view of a portion of the cable connector module in accordance with an exemplary embodiment.

FIG. 6 is a perspective view of the cable in accordance with an exemplary embodiment.

FIG. 7 is a perspective view of the conductor support in accordance with an exemplary embodiment.

FIG. 8 is a perspective view of the bottom ground rake in accordance with an exemplary embodiment.

FIG. 9 is a perspective view of the top ground hood in accordance with an exemplary embodiment.

FIG. 10 is a perspective view of a portion of the cable connector module in accordance with an exemplary embodiment.

FIG. 11 is a top perspective view of a portion of the cable connector module in accordance with an exemplary embodiment.

FIG. 12 is a rear perspective view of a portion of the cable connector module in accordance with an exemplary embodiment.

FIG. 13 is a side view of a portion of the cable connector module in accordance with an exemplary embodiment.

FIG. 14 is a top perspective view of a portion of the cable connector module in accordance with an exemplary embodiment.

FIG. 15 is a cross sectional view of the communication system in accordance with an exemplary embodiment showing the electronic assembly and the circuit board.

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FIG. 16 is a cross sectional view of the communication system in accordance with an exemplary embodiment showing the electronic assembly and the circuit board.

DETAILED DESCRIPTION OF THE INVENTION

FIG. 1 is a front perspective view of a communication system 100 having an electronic assembly 102 in accordance with an exemplary embodiment. FIG. 2 is a rear perspective view of a communication system 100 having an electronic assembly 102 in accordance with an exemplary embodiment. The electronic assembly 102 includes one or more socket assemblies 108 and corresponding cable connector modules 104 (one set shown in FIGS. 1 and 2). The socket assembly 108 is used to electrically connect the corresponding cable connector module 104 to a circuit board 110. An electronic package 106 is electrically connected to the circuit board 110. The cable connector modules 104 are electrically connected to the electronic package 106 through the socket assemblies 108 and the circuit board 110.

One electronic assembly 102 (socket assembly 108 and corresponding cable connector module 104) is shown on one side of the electronic package 106 in FIGS. 1 and 2. However, it should be understood that electronic assemblies 102 may be provided at more than one side, such as all four sides, in alternative embodiments. In various embodiments, a plurality of electronic assemblies 102 may be provided at the side(s) of the electronic package 106. In various embodiments, the cable connector modules 104 are electrical modules using electrical conductors to transmit electrical data signals.

In various embodiments, the electronic package 106 may be an integrated circuit assembly, such as an ASIC. However, the electronic package 106 may be another type of communication component. The electronic package 106 may be mounted directly to the circuit board 110. For example, the electronic package 106 may be soldered to the circuit board 110.

In an exemplary embodiment, compression elements are used to load the cable connector modules 104 against the socket assemblies 108 to electrically connect the cable connector modules 104 to the socket assemblies 108 and to electrically connect the socket assemblies 108 to the circuit board 110. For example, the compression elements may include springs that press the components downward to load the socket assemblies 108 and create mechanical and electrical connections between the cable connector modules 104 and the socket assemblies 108. In various embodiments, the cable connector modules 104 are individually clamped or compressed against the socket assemblies 108 by the compression elements and are thus individually serviceable and removable from the socket assemblies 108.

In an exemplary embodiment, the communication system 100 includes heat dissipating elements (not shown) to dissipate heat from the electronic package 106 and/or the cable connector modules 104.

FIG. 3 is an exploded view of the communication system 100 in accordance with an exemplary embodiment showing the electronic assembly 102 and the circuit board 110. The electronic assembly 102 includes the cable connector module 104 and the socket assembly 108. The socket assembly 108 is used to electrically connect the cable connector module 104 to the circuit board 110. In the illustrated embodiment, the cable connector module 104 is an electrical module having a plurality of cables terminated within the cable connector module 104.

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The circuit board 110 includes a mounting area 114 on an upper surface 116 of the circuit board 110. The socket assembly 108 is coupled to the circuit board 110 at the mounting area 114. The mounting area 114 may be located adjacent to the electronic package 106 (shown in FIG. 1). The circuit board 110 includes board contacts (not shown) at the mounting area 114. The board contacts are arranged in an array, such as in rows and columns. The board contacts may be pads or traces of the circuit board 110. The board contacts may be high speed signal contacts, low speed signal contacts, ground contacts, or power contacts. The board contacts may include pairs of high-speed signal board contacts surrounded by a ring or fence of ground board contacts.

In an exemplary embodiment, the cable connector module 104 includes a connector housing 220 holding a circuit card 302 and a plurality of cable assemblies 300 terminated to the circuit card 302. Each cable assembly 300 includes a cable 310 and a support structure for the cable 310. The support structure is used to couple an end of the cable 310 to the circuit card 302. In an exemplary embodiment, the cable connector module 104 includes a cable holder 304 that holds the cable assemblies 300 relative to the circuit card 302.

The connector housing 220 includes a housing cavity 222 that holds the cable assemblies 300 and the circuit card 302. The connector housing 220 may be a metal shell or cage that receives the cable assemblies 300. Alternatively, the connector housing 220 may be a plastic molded component. In an exemplary embodiment, the housing 220 extends between a top 224 and a bottom 226. The circuit card 302 is provided at the bottom 226. The housing 220 extends between a front 230 and a rear 232. The housing 220 includes sides 234 between the front 230 and the rear 232. The housing includes latching features 235 at the sides 234. The housing 220 includes a pocket 236 at the top 224. The housing 220 may include guide ribs 237 at the top 224 that define the pocket 236. The guide ribs 237 may be located at the front 230 and/or the rear 232 and/or the sides 234. Optionally, at least one of the guide ribs 237 may be approximately centered along the top 224.

In an exemplary embodiment, the socket assembly 108 includes a cage 120 and a socket connector 122 arranged in a cavity 124 of the cage 120. The cavity 124 receives the cable connector module 104 to mate with the socket connector 122 of the socket assembly 108. The cage 120 guides mating of the cable connector module 104 with the socket connector 122.

In an exemplary embodiment, the cage 120 is a stamped and formed cage configured to be stamped and formed from a metal sheet. The cage 120 includes cage walls 240 defining the cavity 124. The cage walls 240 extend between a top 242 and a bottom 244 of the cage 120. The bottom 244 is configured to be coupled to the circuit board 110. In an exemplary embodiment, the cage 120 includes mounting tabs 246 configured to be mounted to the circuit board 110. The mounting tabs 246 may extend from the bottom 244. The mounting tabs 246 may include openings configured to receive fasteners, such as threaded fasteners, used to couple the cage 120 to the circuit board 110. Other types of mounting tabs may be used in alternative embodiments, such as press fit pins, weld tabs, solder tabs, slips, latches, threaded openings, or other types of mounting features. In alternative embodiments, separate securing features may be used to secure the cage 120 to the circuit board 110. In an exemplary embodiment, the cage 120 includes a top opening 148 at the top 242. The top opening 148 is configured to

receive the cable connector module **104**. For example, the cable connector module **104** is top loaded into the cavity **124** through the top opening **148**.

In an exemplary embodiment, the cage **120** includes side walls **250**, **252** and end walls **254**, **256**. The side walls **250**, **252** may be shorter than the end walls **254**, **256**. In the illustrated embodiment, the end wall **256** includes an opening **258**. The opening **258** is configured to receive a portion of the cable connector module **104**, such as the cables. In an exemplary embodiment, the cage **120** includes latching features **260** used for latching coupling to the cable connector module **104**. For example, the latching features **260** interface with the latches **235** of the housing **220** of the cable connector module **104**. The latching features **260** may be deflectable latches. Other types of latching features may be used in alternative embodiments, such as latch openings. The latching features **260** are provided at the side walls **250**, **252** in the illustrated embodiment. However, the latching features **260** may be provided at other locations in alternative embodiments. In an exemplary embodiment, the cage **120** includes openings **262** in alternative embodiments. The openings **262** are provided at the side walls **250**, **252** in the illustrated embodiment. However, the openings **262** may be provided at other locations in alternative embodiments.

In an exemplary embodiment, the electronic assembly **102** includes a spring clip **280** used to couple the cable connector module **104** to the socket connector **122**. The spring clip **280** is configured to engage the cable connector module **104** to hold the cable connector module **104** in the cavity **124** of the cage **120**. In an exemplary embodiment, the spring clip **280** is received in the pocket **236**. The spring clip **280** engages the top **230** and presses against the top **230**. The guide ribs **237** may locate the spring clip **280** relative to the housing **220**. The spring clip **280** presses the cable connector module **104** toward the socket connector **122** to electrically connect the cable connector module **104** to the socket connector **122**.

The spring clip **280** includes latches **282** at first and second sides **284**, **286** of the spring clip **280**. The latches **282** are used to secure the spring clip **280** to the cage **120**. The latches **282** may be coupled to the side walls **250**, **252**, such as to the openings **262**.

The spring clip **280** includes at least one spring beam **290**, such as a pair of spring beams **290**. The spring beams **290** extend between the first side **284** and the second side **286** of the spring clip **290**. The spring beams **290** are connected by connecting beams **292**. The spring beams **290** are curved, such as being curved downward to engage the cable connector module **104**. The spring beams **290** are received in the pocket **236**, such as between the guide ribs **237**. Windows **294** are defined between the spring beams **290**. The windows **284** may receive corresponding guide ribs **237**.

The socket connector **122** includes an array of interposer contacts **200** held together by a substrate **202**. The socket connector **122** may include a frame **204** holding the substrate **202**. The frame **204** may be rectangular. The frame **204** is configured to be coupled to the cage **120**. Additionally, or alternatively, the frame **204** may be coupled to the circuit board **110**. In an exemplary embodiment, the frame **204** includes an opening **206** that receives the substrate **202**. The substrate **202** may fill the opening **206**. The opening **206** may receive a portion of the cable connector module **104**, such as for mating with the interposer contacts **200**. The frame **204** includes frame members **208** defining the opening **206**. In the illustrated embodiment the frame **204** is rectangular having the frame members **208** arranged around the perimeter of the frame **204** (for example, in a rectangular

configuration). The frame **204** is configured to be coupled to the cage **120** and/or the circuit board **110**.

In various embodiments, the substrate **202** is a printed circuit board including the interposer contacts **200** coupled to the printed circuit board. The interposer contacts **200** may be defined by circuits, traces, vias, and the like of the printed circuit board. The interposer contacts **200** may be separate contacts soldered to the printed circuit board.

In other embodiments, the substrate **202** is a film or plate and the interposer contacts **200** are separate contacts which may be held by or coupled to the film. The substrate **202** is manufactured from an insulative material, such as a polyimide material, to electrically isolate the interposer contacts **200** from one another.

In an exemplary embodiment, the interposer contacts **200** are compressible contacts. In various embodiments, the interposer contacts **200** may be stamped and formed contacts, such as dual compression contacts having spring beams at both ends of the contacts and main bodies of the contacts between the spring beams held in the substrate **202**. The interposer contacts **200** may be LGA contacts.

In various embodiments, the interposer contacts **200** are conductive polymer columns. The conductive polymer contacts may be conductive elastomeric connectors having conductive (metallic) particles embedded in an elastomeric material, such as a silicone rubber material. Each interposer contact **200** includes an upper mating interface and a lower mating interface. In various embodiments, the interposer contacts **200** are dual compressible contacts that are compressible at both the upper mating interface and the lower mating interface, such as for mating with the cable connector module **104** and the circuit board **110**, respectively. Optionally, the interposer contacts **200** may be arranged in groups, with each group including a pair of signal contacts surrounded by a ring or fence of ground contacts. The groups are arranged in rows and columns. Other arrangements are possible in alternative embodiments.

FIG. 15 is a cross sectional view of the communication system **100** in accordance with an exemplary embodiment showing the electronic assembly **102** and the circuit board **110**. The cable connector module **104** is coupled to the socket connector **122**. The cable connector module **104** is pressed against the socket connector **122** by the spring clip **280** (FIG. 3) to electrically connect the cable connector module **104** to the socket connector **122**. The cable connector module **104** is pressed downward to compress the interposer contacts **200** of the socket connector **122**. In an exemplary embodiment, the interposer contacts **200** are dual compression contacts that are compressible at the upper interface (for example, with the cable connector module **104**) and at the lower interface (for example, with the circuit board **110**).

Each interposer contact **200** includes an upper contact portion **210** and a lower contact portion **212**. The upper contact portion **210** is compressible. The upper contact portion **210** is configured to be coupled to the corresponding mating contact (not shown) at the bottom of the cable connector module **104**. The lower contact portion **212** is compressible. The lower contact portion **212** is configured to be terminated to the board contact (not shown) at the upper surface of the circuit board **110**. In the illustrated embodiment, the interposer contacts **200** are conductive polymer columns, such as conductive elastomeric connectors. The interposer contacts **200** are held by the substrate **202**. The upper contact portions **210** extend above the substrate **202** for connection to the cable connector module **104**. The lower

contact portions **212** extend below the substrate **202** for connection to the circuit board **110**.

FIG. **16** is a cross sectional view of the communication system **100** in accordance with an exemplary embodiment showing the electronic assembly **102** and the circuit board **110**. The cable connector module **104** is coupled to the socket connector **122**. The cable connector module **104** is pressed against the socket connector **122** by the spring clip **280** (FIG. **3**) to electrically connect the cable connector module **104** to the socket connector **122**. The cable connector module **104** is pressed downward to compress the interposer contacts **200** of the socket connector **122**. In an exemplary embodiment, the interposer contacts **200** are dual compression contacts that are compressible at the upper interface (for example, with the cable connector module **104**) and at the lower interface (for example, with the circuit board **110**).

Each interposer contact **200** includes the upper contact portion **210** and the lower contact portion **212**. The upper and lower contact portions **210**, **212** are compressible. In the illustrated embodiment, the interposer contacts **200** are stamped and formed contacts. The interposer contacts **200** may be LGA contacts. The interposer contacts **200** include upper spring beams at the upper contact portions **210**. The interposer contacts **200** include lower spring beams at the lower contact portions **212**. The upper and lower spring beams compressible. Main bodies of the interposer contacts **200** are held by the substrate **202**. The interposer contacts **200** may include other shapes or features in alternative embodiments. The upper contact portions **210** extend above the substrate **202** for connection to the cable connector module **104**. The lower contact portions **212** extend below the substrate **202** for connection to the circuit board **110**.

With reference back to FIG. **3**, and with additional reference to FIG. **4**, which is a bottom perspective view of the cable connector module **104**, the cable connector module **104** includes the cable assemblies **300** terminated to the circuit card **302**, such as being soldered to the circuit card **302**. The circuit card **302** includes mating pads **388** at a bottom of the circuit card **302**. The mating pads **388** are configured to be mated with corresponding interposer contacts **200** when the cable connector module **104** is plugged into the socket connector **122**. Optionally, the mating pads **388** may be arranged in groups, with each group including a pair of signal pads surrounded by a ring or fence of ground pads. The groups are arranged in rows and columns. Other arrangements are possible in alternative embodiments.

Each cable assembly **300** includes a cable **310** and a support structure for the cable **310**. The support structure is used to couple an end of the cable **310** to the circuit card **302**. The cable connector module **104** may include a heat transfer element (not shown) thermally coupled to the cable assembly **300**, such as to dissipate heat from components on the circuit card **302**.

In an exemplary embodiment, the cable connector module **104** includes a cable holder **304** that holds the cable assemblies **300** relative to the circuit card **302**. The cable connector module **104** includes a connector housing **306** having a cavity **308** that holds the cable assemblies **300** and the circuit card **302**. The connector housing **306** holds the circuit card **302** for mating with the socket assembly **108**. The connector housing **306** may be a metal shell or cage that receives the cable assembly **300**. The connector housing **306** is configured to be coupled to the socket connector **122**, such as to the frame **204**, to position the mating interface of the circuit card **302** relative to the socket connector **122** for mating to the interposer contacts **200**. The cage **120** positions the

connector housing **306**, and thus the circuit card **302**, in the cavity **124** for mating with the socket connector **122**.

The cable holder **304** is coupled to the cables **310** and holds the cables **310** relative to each other within the cavity **308** of the connector housing **306**. The cable holder **304** may be coupled to the circuit card **302** to hold the cables **310** relative to the circuit card **302**. The cable holder **304** provides strain relief for the cables **310**. In alternative embodiments, the cable assembly **300** may be provided without the cable holder **304**. Rather, the cables **310** may be unsupported or freely arranged in the connector housing **306**. The circuit card **302** is coupled to the connector housing **306** to position the mating interface of the circuit card **302** for mating with the socket assembly **108**. The cable holder **304** is coupled to the connector housing **306** to position the cables **310** relative to the connector housing **306**.

FIG. **5** is a top perspective view of a portion of the cable connector module **104** in accordance with an exemplary embodiment. FIG. **5** shows a plurality of the cable assemblies **300** terminated to the circuit card **302**. The cable assemblies **300** are supported by the cable holder **304**. The connector housing **306** (shown in FIG. **3**) is removed to illustrate the cable assemblies **300** and the circuit card **302**.

In the illustrated embodiment, the cable assemblies **300** are stacked in multiple rows, such as three or more rows. In various embodiments, the cable connector module **104** may include at least thirty-six (36) cable assemblies **300** terminated to the circuit card **302** in an area of approximately 30 mm by 20 mm. For example, the cable assemblies **300** may be arranged in a 4×9 matrix. However, the cable connector module **104** may include greater or fewer cable assemblies **300** in the termination area in alternative embodiments. The termination area may be larger or smaller in alternative embodiments, which may include greater or fewer cable assemblies **300** in alternative embodiments. The cable assemblies **300** may be grouped together, such as in three or more groups (only one group is illustrated in FIG. **5**). Each group of cable assemblies **300** are terminated to a pad or section of the circuit card.

The circuit card **302** includes an upper surface **380** and a lower surface **382** opposite the upper surface **380**. In an exemplary embodiment, the circuit card **302** includes contact pads **384** at the upper surface **380** configured to be electrically connected to the signal conductors of cables **310**. The contact pads **384** are circuits of the circuit card **302** and may be connected to plated vias extending through the circuit card **302** to the lower surface **382**. The contact pads **384** may be arranged in groups, such as pairs. In the illustrated embodiment, the contact pads **384** are all provided on the upper surface **380**. However, in alternative embodiments, the contact pads **384** may additionally be provided on the lower surface **382**. Optionally, the contact pads **384** may be arranged in a ground-signal-signal-ground arrangement. In the illustrated embodiment, the contact pads **384** are provided in multiple rows along the circuit card **302**. The rows are aligned parallel to the front and the rear of the circuit card **302**. The circuit card **302** is densely populated with the contact pads **384** to allow a large number of cables **310**, and thus a large number of signal lines, to be electrically connected to the circuit card **302**.

In an exemplary embodiment, the circuit card **302** includes ground vias **386** at the upper surface **380**. The ground vias **386** are located proximate to the contact pads **384**. The ground vias **386** are electrically connected to the ground plane. The cable assemblies **300** are coupled to the corresponding ground vias **386**.

The circuit card 302 includes mating pads 388 (shown in FIG. 4) at the lower surface 382 configured to be electrically connected to corresponding interposer contacts 200 of the socket assembly 108 (both shown in FIG. 3). The mating pads 388 are electrically connected to corresponding contact pads 384 through plated vias or other circuits of the circuit card 302. In an exemplary embodiment, the mating pads 388 are all provided on the lower surface 382. However, the mating pads 388 may be provided on both the upper surface 380 and the lower surface 382 in alternative embodiments, such as when the circuit card 302 is a pluggable card configured to be plugged into a card slot of a receptacle connector.

In an exemplary embodiment, each cable assembly 300 includes the cable 310, a conductor support 312 coupled to the end of the cable 310, and a ground structure 314 used to electrically connect the cable 310 to the circuit card 302. In various embodiments, the ground structure 314 is a ground clip and may be referred to hereinafter as a ground clip 314. The ground clip 314 is coupled to the end of the cable 310. The ground clip 314 and the conductor support 312 support the cable 310 relative to the circuit card 302. For example, the ground clip 314 and the conductor support 312 hold the cable 310 at an angle relative to the circuit card 302 to allow the cable 310 to lift off of and away from the circuit card 302, rather than laying flat or parallel to the circuit card 302. Such an arrangement allows tighter packaging of the cable assemblies 300, such as to increase the density of the cable connector module 104 for a given footprint of the circuit card 302. Other types of ground structures 314 may be used in alternative embodiments to electrically connect the cable 310 to the circuit card 302, such as a bus bar, a crimp barrel, a ground cage, or ground hood, and the like. The ground structure 314 may be a stamped and formed structure.

The ground clip 314 is used to mechanically and electrically connect the cable 310 to the circuit card 302. In an exemplary embodiment, the ground clip 314 is a multi-piece structure. For example, the ground clip 314 may include multiple stamped and formed pieces that are stamped and formed separately from each other and then coupled together, such as being welded together to form the ground clip 314. In the illustrated embodiment, the ground clip 314 includes a bottom ground rake 316 and a top ground hood 318. Optionally, the top ground hood 318 of multiple cable assemblies 300, such as the cable assemblies within each row, may be ganged or integrated together as a single top ground hood. However, in alternative embodiments, each cable assembly 300 may include a separate top ground hood 318.

In an exemplary embodiment, the ground clip 314 is used to direct the cable 310 away from the circuit card 302, such as at a predetermined cable exit angle. For example, the ground clip 314 may hold the cable 310 at a cable exit angle that is non-parallel to the circuit card 302 and non-perpendicular to the circuit card 302. In various embodiments, the ground clip 314 may hold the cable 310 at a cable exit angle of between 300 and 60°. Optionally, the ground clip 314 may hold the cable 310 at a cable exit angle of approximately 45°. The ground clip 314 holds the cable 310 at an angle to allow tight spacing of the cable assemblies 300. For example, the contact pads on the circuit card 302 may be more densely populated (for example, closer spacing) by forcing the cable 310 to exit at an angle from the circuit card 302 as compared to conventional cable connector modules having the cables oriented generally parallel to the circuit card for termination to the circuit card.

In an exemplary embodiment, the ground clip 314 is used to electrically connect to the cable 310, such as to improve electrical performance of the cable connector module 104. For example, the ground clip 314 may reduce excess insertion loss and cross talk due to tighter control of electromagnetic fields at the termination area. The ground clip 314 may electrically connect to the cable 310 at multiple locations. For example, the ground clip 314 may electrically connect at the top, the bottom and both sides to provide nearly circumferential connection between the cable 310 and the ground clip 314. The ground clip 314 positions the cable 310 to have a short ground return path between the cable 310 and the circuit card 302 for improved electrical characteristics.

Each cable 310 extends between the ground clip 314 and the cable holder 304. The cables 310 transition between the ground clips 314 and the cable holder 304. For example, the cables 310 lift off of the circuit card 302 immediately rearward of the termination area and extend rearward to the cable holder 304. The cables 310 are stacked in rows in the cable holder 304. In an exemplary embodiment, the cable holder 304 is coupled to the circuit card 302 to hold the cables 310 relative to the circuit card 302. The cable holder 304 provides strain relief for the cables 310. The cables 310 are exposed to air between the cable holder 304 and the ground clips 314.

The circuit card 302 is coupled to the connector housing 306 to position the mating interface of the circuit card 302 for mating with the socket assembly 108. The cable holder 304 is coupled to the connector housing 306 to position the cables 310 relative to the connector housing 306. In alternative embodiments, the cable assembly 300 may be provided without the cable holder 304. Rather, the cables 310 may be unsupported or freely arranged in the connector housing 306.

In an exemplary embodiment, the cable holder 304 includes cable supports 390 arranged in a cable support stack 392. The cable supports 390 hold the cables 310 at elevated positions above the circuit card 302. For example, the cable supports 390 hold the cables 310 at different heights above the upper surface 380 of the circuit card 302. In an exemplary embodiment, each cable support 390 extends between a top 391 and a bottom 393. The cable supports 390 are stacked bottom-to-top on top of each other. The cable supports 390 may include locating features 395 for locating the cable supports 390 within the cable support stack 392. For example, the locating features 395 may be posts and openings where the posts are received in openings to position and/or secure the cable supports 390 together. In alternative embodiments, the cable holder 304 may be a single piece structure that holds the cables 310 rather than multiple, stacked cable supports 390.

In an exemplary embodiment, the cable holder 304 includes cable channels 394 therethrough that receive corresponding cables 310. The cable channels 394 are arranged in multiple rows. In the illustrated embodiment, the cable supports 390 include the cable channels 394 that receive corresponding cables 310. The cable channels 394 may be open at the top 391 and the bottom 393 of each cable support 390 to receive corresponding cables 310 along both the top 391 and the bottom 393 of the cable support 390. For example, the cable channels 394 include upper cable channels 394a at the top 391 and lower cable channels 394b at the bottom 393. Optionally, the cables 310 may be sandwiched between the cable supports 390. Alternatively, the cable channels 394 may be contained within the cable

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supports 390, such as with each cable support 390 entirely circumferentially surrounding the corresponding cable channels 394.

In an exemplary embodiment, the cables 310 extend forward of the cable supports 390 to the ground clips 314. The cables 310 may be preformed into a particular shape between the cable supports 390 and the ground clips 314. For example, bends may be formed in the cables 310 at predetermined locations (lengths forward of the cable supports 390 and/or rearward from the ends of the cables 310). The cables 310 may be shape-retaining to retain the preformed bend between the cable holder 304 and the ground clip 314. Each row of cables 310 may have different shapes (for example, bends at different locations).

In the illustrated embodiment, the cables 310 each have a horizontal portion immediately forward of the cable supports 390 and an angled portion between the horizontal portion and the ground clip 314. In various embodiments, the angled portion is angled between approximately 150° and 120° relative to the horizontal portion. The angled portion may be angled at approximately 135° relative to the horizontal portion. In an exemplary embodiment, each cable 310 includes an end portion 311, a bend portion 313 rearward of the end portion 311, and a support portion 315 rearward of the bend portion 313. The cable 310 is bent and non-linear in the bend portion 313. The cable 310 is generally straight (linear) along the end portion 311 and the support portion 315. The end portion 311 extends between the bend portion 313 and the ground clip 314 and the end of the cable 310. The support portion 315 extends between the bend portion 313 and the cable holder 304. The support portion 315 passes through the cable holder 304. In an exemplary embodiment, the cables 310 are arranged in an inner row, an outer row, and at least one intermediate row. The cables 310 in the inner row are terminated to the circuit card 302 closest to the cable holder 304. The cables 310 in the outer row are terminated to the circuit card 302 furthest from the cable holder 304. The end portions 311 of the cables 310 in the inner row are shorter than the end portions 311 of the cables 310 in the outer row. The support portions 315 of the cables 310 in the inner row are shorter than the support portions 315 of the cables 310 in the outer row. The bend portions 313 of the cables 310 in the inner row are located closer to the cable holder 304 than the bend portions 313 of the cables 310 in the outer row. The bend portions 313 of the cables 310 in the inner row are located at an elevation lower than the bend portions 313 of the cables 310 in the outer row.

In various embodiments, the cables 310 may be surrounded by epoxy or hot melt forward of the cable supports 390 as a strain relief for the cables 310. The cable supports 390 may form a dam to form the strain relief against. In other embodiments, the cables 310 may be in open air forward of the cable supports 390, such as to improve signal integrity at the cable termination area compared to embodiments having epoxy or hot melt around the cables 310. The cable supports 390 may form a dam to allow epoxy or hot melt to form around the cables 310 rearward of the cable supports 390, such as to enhance the strain relief provided by the cable supports 390.

FIG. 6 is a perspective view of the cable 310 in accordance with an exemplary embodiment. The cable 310 includes at least one signal conductor and a shield structure providing electrical shielding for the at least one signal conductor. In an exemplary embodiment, the cables 310 are twin-axial cables. For example, each cable 310 includes a first signal conductor 320 and a second signal conductor

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322. The signal conductors 320, 322 carry differential signals. The cable 310 includes an insulator 324 surrounding the signal conductors 320, 322 and a cable shield 326 surrounding the insulator 324. In various embodiments, the insulator 324 includes a single core surrounding both signal conductors 320, 322. In other various embodiments, the insulator 324 is a dual core insulator having a first dielectric element surrounding the first signal conductor 320 and a second dielectric element surrounding the second signal conductor 322. The cable shield 326 provides circumferential shielding around the signal conductors 320, 322. The cable 310 includes a cable jacket 328 surrounding the cable shield 326. In various embodiments, the cable 310 includes one or more drain wires 329 electrically connected to the cable shield 326, such as a pair of drain wires 329 extending along opposite sides of the cable 310, such as between the cable shield 326 and the cable jacket 328. The drain wire(s) 329 are configured to be terminated to the circuit card 302, such as being soldered to contact pads of the circuit card 302. In an exemplary embodiment, the drain wires 329 are configured to be electrically connected to the ground clip 314 (shown in FIG. 5).

In an exemplary embodiment, at an end of the cable 310, the cable jacket 328, the cable shield 326, and the insulator 324 may be removed (e.g., stripped) to expose portions of the signal conductors 320, 322. Exposed portions 321, 323 of the signal conductors 320, 322 extend forward from an end 325 of the insulator. The exposed portions 321, 323 are configured to be mechanically and electrically coupled (e.g., soldered) to corresponding contact pads 384 on the circuit card 302 (shown in FIG. 5). The exposed portions 321, 323 may be bent, such as bent inward toward each other (distance between reduced for tighter coupling and smaller trace spacing) and/or may be bent to extend along the surface of the circuit card 302 to terminate to the contact pads 384.

FIG. 7 is a perspective view of the conductor support 312 in accordance with an exemplary embodiment. Optionally, multiple conductor supports 312 may be combined to form a unitary structure. For example, FIG. 7 illustrates the conductor support for three cables 310 as a unitary structure. It should be understood that a single conductor support 312 for a single cable 310 may be utilized in alternative embodiments. The conductor support 312 includes a dielectric body used to hold the exposed portions 321, 323 of the signal conductors 320, 322 (shown in FIG. 6). The conductor support 312 electrically isolates the signal conductors 320, 322 from each other and from the ground clip 314 (shown in FIG. 5).

The conductor support 312 includes conductor channels 330 that receive the signal conductors 320, 322. The conductor channels 330 extend between a front and a rear of the conductor support 312. The conductor channels 330 position the signal conductors 320, 322 relative to each other. The conductor channels 330 may pass straight through the conductor support 312 between the front and the rear. However, in alternative embodiments, the conductor channels 330 may be curved or angled to change relative positions of the signal conductors 320, 322 between the front and the rear. For example, the conductor channels 330 may be closer together at the front and further apart at the rear. The conductor channels 330 may be open at the top or at the bottom of the conductor support 312 to receive the signal conductors 320, 322 through the top side or the bottom side of the conductor support 312. Alternatively, the signal conductors 320, 322 may be fed into the conductor channels 330 through the rear of the conductor support 312.

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The conductor support 312 includes a rear wall 332 at the rear of the conductor support 312. The rear wall 332 is configured to face the end 325 of the insulator 324 (shown in FIG. 6). The rear wall 332 may abut against the insulator 324. The conductor support 312 includes a nose cone 334 at a front portion 336 of the conductor support 312. The nose cone 334 at the front portion 336 is configured to be received in the ground clip 314. The conductor channels 330 pass through the front portion 336. In an exemplary embodiment, side walls 338 of the conductor support 312 may be angled inward from the rear wall 332 to the nose cone 334.

In various embodiments, conductor supports 312 of multiple cable assemblies 300 may be molded together as a unitary structure. By molding the conductor supports 312 together, the spacing between the cable assemblies 300 may be controlled by the conductor supports 312.

FIG. 8 is a perspective view of the bottom ground rake 316 in accordance with an exemplary embodiment. Optionally, multiple bottom ground rakes 316 may be combined to form a unitary structure. For example, FIG. 8 illustrates the bottom ground rake for three cables 310 as a unitary structure. It should be understood that a single bottom ground rake 316 for a single cable 310 may be utilized in alternative embodiments. The bottom ground rake 316 is used as a mechanical and electrical connector between the cable 310 and the circuit card 302 (shown in FIG. 5). The bottom ground rake 316 is manufactured from a conductive material, such as a metal material. In an exemplary embodiment, the bottom ground rake 316 is stamped and formed from a metal plate into a shape configured to mechanically and electrically connect the cable 310 to the circuit card 302. The bottom ground rake 316 is configured to be electrically connected to the cable shield 326 of the cable 310. The bottom ground rake 316 is configured to be electrically connected to the circuit card 302.

The bottom ground rake 316 includes a support wall 340 used to support the cable 310. The bottom ground rake 316 includes a lower grounding tab 342 extending from the support wall 340. The lower grounding tab 342 is configured to be electrically connected to the cable 310, such as to a lower portion of the cable shield 326. The lower grounding tab 342 includes a generally planar inner surface that faces the lower portion of the cable shield 326. The lower grounding tab 342 has a large surface area for electrical connection with the cable shield 326. In various embodiments, the inner surface of the lower grounding tab 342 may be directly coupled to the lower portion of the cable shield 326 to create a DC electrical connection with the cable shield 326. In other embodiments, the inner surface of the lower grounding tab 342 may be spaced apart from, but located in close proximity to, the lower portion of the cable shield 326 to create a capacitive electrical connection between the lower grounding tab 342 and the cable shield 326.

The bottom ground rake 316 includes side connecting tines 344 configured to be electrically connected to the sides of the cable 310. In various embodiments, the side connecting tines 344 may be directly connected to the cable shield 326 at the sides of the cable 310, such as being soldered or compression coupled to the cable shield 326 at the sides of the cable 310. Alternatively, the side connecting tines 344 may be located in close proximity to the sides of the cable shield 326 to create a capacitive electrical connection between the side connecting tines 344 and the cable shield 326. However, in alternative embodiments, the side connecting tines 344 may be electrically connected to the sides of the cable 310 via the drain wires 329. For example, the side connecting tines 344 may be drain wire tines and may

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be referred to hereinafter as drain wire tines 344. The drain wire tines 344 extending from the support wall 340 for electrical connection to the drain wires 329 of the cable 310. For example, the drain wire tines 344 may be provided at both sides of the support wall 340 to connect with both drain wires 329. In an exemplary embodiment, drain wire slots 346 are defined between pairs of the drain wire tines 344 that receive the drain wires 329. The drain wire tines 344 may be connected to the drain wires 329 by an interference fit. Alternatively, the drain wire tines 344 may be soldered to the drain wires 329. The drain wires 329 create direct electrical paths between the bottom ground rake 316 and the cable shield 326. For example, the drain wires 329 are directly coupled to (DC electrical connection) the drain wire tines 344 and to the cable shield 326.

The bottom ground rake 316 includes one or more mounting tabs 348 extending from the support wall 340. The mounting tabs 348 are used to mount the bottom ground rake 316 to the circuit card 302. In the illustrated embodiment, the mounting tabs 348 are compliant pins, such as eye-of-the-needle pins, configured to be press fit into plated vias of the circuit card 302. In alternative embodiments, the mounting tabs 348 may be solder tabs configured to be soldered to the circuit card 302.

In an exemplary embodiment, the support wall 340 includes a lower panel 350 and a support panel 352. The lower panel 350 and the support panel 352 form a pocket that receives the conductor support 312 and the end of the cable 310. The lower panel 350 defines a base of the support wall 340 that is configured to be mounted to the circuit card 302. For example, the lower panel 350 is configured to rest on the upper surface of the circuit card 302. The support panel 352 extends forward and upward at an angle from the lower panel 350. The support panel 352 supports the front of the conductor support 312 and the end of the cable 310. The lower panel 350 supports the bottom of the conductor support 312.

In an exemplary embodiment, the lower grounding tab 342 extends rearward and upward from the lower panel 350 at an angle, which may define the cable exit direction for the cable 310 from the bottom ground rake 316. For example, the lower grounding tab 342 may be angled transverse (for example, non-parallel) to the lower panel 350 to extend along the cable 310. In various embodiments, the lower grounding tab 342 may be angled at between 30° and 60° relative to the lower panel 350 (horizontal), such as approximately 45°. The mounting tabs 348 extend rearward from the lower panel 350 for connection to the circuit card 302. The drain wire tines 344 extend forward from the lower panel 350 and/or the support panel 352.

In an exemplary embodiment, the support panel 352 is angled transverse (for example, non-parallel) relative to the lower panel 350. For example, the support panel 352 may be angled at between 30° and 60° relative to the lower panel 350, such as approximately 45°. The angles of the panels 350, 352 control the cable exit angle from the bottom ground rake 316 and thus the circuit card 302. For example, the plane defined by the support panel 352 defines the angle of the end 325 of the insulator 324 of the cable 310. The cable 310 extends from the bottom ground rake 316 in a cable exit direction that is perpendicular to the plane of the support panel 352.

In an exemplary embodiment, the support panel 352 includes a window 354 therethrough. The window 354 is configured to receive the signal conductors 320, 322. The window 354 is configured to receive the nose cone 334 at the front portion 336 of the conductor support 312. Optionally,

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the window 354 is open at a top of the support panel 352. The drain wire tines 344 and the drain wire slots 346 are located on opposite sides of the support panel 352.

FIG. 9 is a perspective view of the top ground hood 318 in accordance with an exemplary embodiment. Optionally, multiple top ground hoods 318 may be combined to form a unitary structure. For example, FIG. 9 illustrates the top ground hood for three cables 310 as a unitary structure. It should be understood that a single top ground hood 318 for a single cable 310 may be utilized in alternative embodiments. The top ground hood 318 is used as a mechanical and electrical connector between the cable 310 and the circuit card 302 (shown in FIG. 5). The top ground hood 318 is manufactured from a conductive material, such as a metal material. In an exemplary embodiment, the top ground hood 318 is stamped and formed from a metal plate into a shape configured to mechanically and electrically connect the cable 310 to the circuit card 302. The top ground hood 318 is configured to be electrically connected to the cable shield 326 of the cable 310. The top ground hood 318 is configured to be electrically connected to the circuit card 302.

The top ground hood 318 includes a cover 360 used to shield the termination area of the cable 310 to the circuit card 302. The cover 360 has an inner surface 362 that defines a shield pocket 364. The exposed portions of the signal conductors extend into the shield pocket 364 for termination to the circuit card 302. In an exemplary embodiment, the cover 360 includes an upper cover panel 366 above the shield pocket 364 and a front cover panel 368 forward of the shield pocket 364.

The top ground hood 318 includes an upper grounding tab 370 extending from the cover 360. The upper grounding tab 370 is configured to be electrically connected to the cable 310, such as to an upper portion of the cable shield 326. The upper grounding tab 370 includes a generally planar inner surface that faces the upper portion of the cable shield 326. The upper grounding tab 370 has a large surface area for electrical connection with the cable shield 326. In various embodiments, the inner surface of the upper grounding tab 370 may be directly coupled to the upper portion of the cable shield 326 to create a DC electrical connection with the cable shield 326. In other embodiments, the inner surface of the upper ground tab 370 may be spaced apart from, but located in close proximity to, the upper portion of the cable shield 326 to create a capacitive electrical connection between the upper grounding tab 370 and the cable shield 326.

In an exemplary embodiment, the upper grounding tab 370 extends rearward and upward from the upper cover panel 366 at an angle, which may define the cable exit direction for the cable 310 from the top ground hood 318. For example, the upper grounding tab 370 may be angled transverse (for example, non-parallel) to the upper cover panel 366 (horizontal) to extend along the cable 310. In various embodiments, the upper grounding tab 370 may be angled at between 30° and 60° relative to the upper cover panel 366, such as approximately 45°.

The top ground hood 318 includes one or more mounting tabs 372 extending from the cover 360. The mounting tabs 372 are used to mount the top ground hood 318 to the circuit card 302. In the illustrated embodiment, the mounting tabs 372 are compliant pins, such as eye-of-the-needle pins, configured to be press fit into plated vias of the circuit card 302. In alternative embodiments, the mounting tabs 372 may be solder tabs configured to be soldered to the circuit card 302. In the illustrated embodiment, the mounting tabs 372

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extend from the bottom edge of the front cover panel 368 for connection to the circuit card 302.

In an exemplary embodiment, the top ground hood 318 including one or more connecting tabs 374 extending from the cover 360. The connecting tabs 374 are used to mechanically and electrically connect the top ground hood 318 to the bottom ground rake 316. In the illustrated embodiment, the connecting tabs 374 extend from the sides of the cover 360. The connecting tabs 374 are configured to be soldered or welded to the bottom ground rake 316, such as to the support wall 340 or drain wire tines 344.

FIG. 10 is a perspective view of a portion of the cable connector module 104 in accordance with an exemplary embodiment. FIG. 10 shows the cable connector module 104 during an intermediate stage of assembly. During assembly, the cable assemblies 300 are coupled to the circuit card 302. In various embodiments, the bottom ground rake 316 and/or the conductor support 312 may be coupled to the end of the cable 310 prior to coupling the bottom ground rake 316 and the conductor support 312 to the circuit card 302. However, in alternative embodiments, the bottom ground rake 316 and the conductor support 312 may be coupled to the circuit card 302 prior to coupling the end of the cable 310 to the bottom ground rake 316 and the conductor support 312.

During assembly, the exposed portions 321, 323 of the signal conductors 320, 322 are coupled to the conductor support 312. The signal conductors 320, 322 are loaded into the conductor channels 330. For example, the conductor channels 330 may be open at the top of the conductor support 312 such that the signal conductors 320, 322 may be loaded into the conductor channels 330 from above. The rear wall 332 faces, and may abut against, the end 325 of the insulator 324. In an exemplary embodiment, the cable 310 extends away from the conductor support 312 in a cable exit direction. The angle of the rear wall 332 (relative to the circuit card 302) controls the cable exit direction relative to the circuit card 302. For example, the cable exit direction may be perpendicular to the rear wall 332. The conductor support 312 may be used to support the drain wires 329. The dielectric material of the conductor support 312 controls impedance along the signal paths. In various embodiments, the conductor support 312 is made of low loss material to decrease insertion loss along the signal paths.

In various embodiments, the conductor support 312 may be coupled to the bottom ground rake 316 of the ground clip 314. For example, the bottom ground rake 316 may position and support the conductor support 312 to position the conductor support 312 relative to the circuit card 302 to control the position and orientation of the cable 310 (for example, to control the cable exit direction). Optionally, the conductor support 312 may be pre-formed separate from the bottom ground rake 316, such as being molded separate from the bottom ground rake 316. The conductor support 312 is then coupled to the bottom ground rake 316, such as being press-fit into the ground clip and held by an interference fit. The conductor support 312 may be secured to the bottom ground rake 316 using adhesive, fasteners, clips or other securing means.

The signal conductors 320, 322 may be loaded into the conductor support 312 prior to coupling to the bottom ground rake 316. Alternatively, the signal conductors 320, 322 may be loaded into the conductor support 312 after the conductor support 312 is coupled to the bottom ground rake 316.

In alternative embodiments, the conductor support 312 may be formed in place on the bottom ground rake 316 rather than being separately manufactured and then

assembled with the bottom ground rake **316**. For example, the conductor support **312** may be overmolded over a portion of the bottom ground rake **316**. The conductor support **312** is molded to the bottom ground rake **316** to fix the position of the conductor support **312** relative to the bottom ground rake **316**.

In various embodiments, the conductor support **312** of multiple cable assemblies **300** may be integrated as a unitary, monolithic structure. For example, the conductor supports **312** may be co-molded together, which controls spacing and relative positioning of the cables **310**. In various embodiments, the conductor supports **312** may be overmolded over the bottom ground rakes **316** to control relative positioning of the bottom ground rakes **316** of multiple cable assemblies **300**.

When assembled, the exposed portions **321**, **323** of the conductors **320**, **322** extend forward of the conductor support **312**, such as for termination to the circuit card **302**. The drain wires **329** pass through the drain wire slots **346** forward of the bottom ground rake **316** for termination to the circuit card **302**. The drain wires **329** may be held in the drain wire slots **346** by an interference fit to electrically connect the drain wires **329** to the bottom ground rake **316**. Alternatively, the drain wires **329** may be soldered to the drain wire tines **344** to electrically connect the drain wires **329** to the bottom ground rake **316**. The drain wires **329** connect the bottom ground rake **316** to the cable shield **326**, such as to opposite sides (right side and left side) of the cable **310**.

During assembly, the mounting tabs **348** of the bottom ground rake **316** are coupled to the circuit card **302**. For example, the mounting tabs **348** may be press-fit into the ground vias **386**. Alternatively, the mounting tabs **348** may be soldered to corresponding circuits of the circuit card **302**.

During assembly, the exposed portions **321**, **323** of the signal conductors **320**, **322** are coupled to the corresponding contact pads **384** at the upper surface **380** of the circuit card **302**. For example, the signal conductors **320**, **322** are soldered to the contact pads **384**. Similarly, the ends of the drain wires **329** are coupled to the corresponding contact pads **384** of the circuit card **302**. For example, the drain wires **329** are soldered to the contact pads **384** to connect to a ground plane at the upper surface **380** of the circuit card **302**. The drain wires **329** extend along (for example, parallel to and spaced apart from) the exposed portions **321**, **323**. The drain wires **329** are located between the pairs of exposed portions **321**, **323** to provide shielding between the pairs of signal conductors **320**, **322**. The signal conductors **320**, **322** transition from the end of the insulator **324**, through the conductor support **312**, to the circuit card **302**. The exposed portions **321**, **323** of the signal conductors **320**, **322** are bent at an angle relative to the cable axis to transition to the circuit card **302**.

FIG. **11** is a top perspective view of a portion of the cable connector module **104** in accordance with an exemplary embodiment. FIG. **12** is a rear perspective view of a portion of the cable connector module **104** in accordance with an exemplary embodiment. FIGS. **11** and **12** show the top ground hood **318** coupled to the bottom ground rake **316** and the circuit card **302**. In an exemplary embodiment, the top ground hood **318** is assembled after the bottom ground rake **316** and the cables **310** are terminated to the circuit card **302**.

During assembly, the mounting tabs **372** extending from the cover **360** are coupled to the circuit card **302**. For example, the mounting tabs **372** are press-fit into the ground vias **386**. Alternatively, the mounting tabs **372** may be soldered to corresponding circuits of the circuit card **302**.

During assembly, the connecting tabs **374** extending from the cover **360** are coupled to the bottom ground rake **316**. For example, the connecting tabs **374** are soldered or welded to the bottom ground rake **316**, such as to the support wall **340**. The support wall **340** and/or the drain wire tines **344** support the rear end of the top ground hood **318**, such as to position the top ground hood **318** relative to the conductors **320**, **322**. In an exemplary embodiment, the connecting tabs **374** are connected to the drain wire tines **344**, such as being welded to the drain wire tines **344**.

FIG. **13** is a side view of a portion of the cable connector module **104** in accordance with an exemplary embodiment. The cable assemblies **300** are coupled to the circuit card **302**. The ground clip **314** is coupled to the circuit card **302**. The ground clip **314** defines the cable exit angle from the circuit card **302**. The ground clip **314** provides electrical shielding for the cable **310** at the termination zone between the cable **310** and the circuit card **302**. The ground clip **314** creates an electrical path between the circuit card **302** and the cable shield **326** of the cable **310**. For example, the ground clip includes multiple electrical interfaces with the cable shield **326**, such as at the upper portion, the lower portion, and both the right and left side portions of the cable shield **326**. The electrical interfaces between the ground clip **314** and the cable **310** may be made via solderless connections. For example, the upper and lower connections may be made by direct, interference fit connections or capacitive coupling between the ground clip **314** and the cable shield and the right and left side connections may be made by interference fit connections between the ground clip **314** and the drain wires **329**. The multiple connection points are provided around the cable shield **326**, such as on all four sides of the cable shield **326** to efficiently connect the ground clip **314** and the cable shield **326** allowing efficient operation at high frequencies, such as between DC and 67 GHz. The ground structure allows efficient high-speed operation for the system, such as at 224 Gbps.

The circuit card **302** extends between the upper surface **380** and the lower surface **382**. The contact pads **384** are provided at the upper surface **380**. The mating pads **388** are provided at the lower surface **382**. In an exemplary embodiment, the contact pads **384** include both signal contact pads **384a** (FIG. **11**) and ground contact pads **384b** (FIG. **11**). The ground contact pads **384b** may be discrete contact pads. In other various embodiments, the ground contact pads **384b** may be defined by a ground plane **383** at the upper surface **380**. The contact pads **384** are connected to corresponding mating pads **388** by plated vias **385**. The plated vias **385** extend through the circuit card **302** between the upper surface **380** and the lower surface **382**. The plated vias **385** may be signal vias **385a** and ground vias **385b**. The signal vias **385a** electrically connect the signal contact pads **384a** and the signal mating pads **388a**. The ground vias **385b** electrically connect the ground contact pads **384b** and the ground mating pads **388b**. In an exemplary embodiment, the signal contact pads **384a** and the signal mating pads **388a** are arranged in pairs. The ground contact pads **384b**, the ground mating pads **388b**, and the ground vias **385b** surround the signal contact pads **384a**, the signal mating pads **388a**, and the signal vias **385a**, such as forming a ring or fence around the signal pairs.

In an exemplary embodiment, the contact pads **384** and the mating pads **388** are electrically connected only by the plated vias **385**. For example, the contact pads **384** and the signal mating pads **388** are electrically connected without circuit traces routed on other layers of the circuit card **302**. In an exemplary embodiment, the contact pads **384** are

vertically aligned with the corresponding mating pads **388**. The plated vias **385** pass vertically through the circuit card **302** along via axes **387**. The contact pads **384** and the signal mating pads **388** are coincident with the via axes **387**. The plated vias **385** are oriented perpendicular to the upper surface **380** and the lower surface **382**. The vias **385** have the shortest length between the contact pad **384** and the corresponding mating pad **388** for high speed signaling through the circuit card **302**. The plated vias **385** extend the entire height of the circuit card **302** from the upper surface **380** to the lower surface **382** to connect the contact pads **384** and the mating pads **388**.

When assembled, the bottom ground rake **316** and the top ground hood **318** are coupled to the circuit card **302** and form a cable pocket **376** that extends along a cable exit axis **378**. The end of the cable **310** is received in the cable pocket **376**. The cable **310** extends away from the circuit card **302** in a cable exit direction along the cable exit axis **378**. The conductor support **312** is received in the cable pocket **376** and supports the end of the cable **310**. The cable pocket **376** is defined between inner surfaces of the lower grounding tab **342** and the upper grounding tab **370**. The inner surfaces may be planar surfaces oriented parallel to each other. The lower grounding tab **342** and the upper grounding tab **370** are spaced apart from each other to receive the cable **310** therebetween. The lower grounding tab **342** and the upper grounding tab **370** extend parallel to the cable exit axis **378**. The lower grounding tab **342** and the upper grounding tab **370** position the cable **310** and locate the cable **310** along the cable exit axis **378**. The lower grounding tab **342** supports the cable **310** from below. The upper grounding tab **370** supports the cable **310** from above.

In various embodiments, the lower grounding tab **342** directly engages the cable shield **326** of the cable **310** to electrically connect (DC electrical connection) the bottom ground rake **316** to the cable shield **326**. Additionally, or alternatively, the lower grounding tab **342** is capacitively coupled to the lower portion of the cable shield **326**. For example, the lower grounding tab **342** is closely positioned relative to the cable shield **326** but does not physically contact the cable shield **326** (small separation distance). The lower grounding tab **342** may be positioned at most 50 microns apart from the cable shield **326** to create a strongly capacitively coupled connection between the lower grounding tab **342** and the lower portion of the cable shield **326**. The large surface area of the lower grounding tab **342** provides an efficient capacitive connection between the lower grounding tab **342** and the cable shield **326**. A ground return path is defined between the bottom ground rake **316** and the cable shield **326** through the DC electrical connection or the capacitive connection between the lower grounding tab **342** and the lower portion of the cable shield **326**.

In various embodiments, the upper grounding tab **370** directly engages the cable shield **326** of the cable **310** to electrically connect (DC electrical connection) the top ground hood **318** to the cable shield **326**. Additionally, or alternatively, the upper grounding tab **370** is capacitively coupled to the upper portion of the cable shield **326**. For example, the upper grounding tab **370** is closely positioned relative to the cable shield **326** but does not physically contact the cable shield **326** (small separation distance). The upper grounding tab **370** may be positioned at most 50 microns apart from the cable shield **326** to create a strongly capacitively coupled connection between the upper grounding tab **370** and the upper portion of the cable shield **326**. The large surface area of the upper grounding tab **370** provides an efficient capacitive connection between the

upper grounding tab **370** and the cable shield **326**. A ground return path is defined between the top ground hood **318** and the cable shield **326** through the DC electrical connection or the capacitive connection between the upper grounding tab **370** and the upper portion of the cable shield **326**.

In an exemplary embodiment, the lower grounding tab **342** and the upper grounding tab **370** are angled transverse relative to the circuit card **302**. For example, the lower grounding tab **342** and the upper grounding tab **370** may be angled between 30° and 60° relative to the (horizontal) circuit card **302**. Optionally, the lower grounding tab **342** and the upper grounding tab **370** may be angled at approximately 45°. The lower grounding tab **342** and the upper grounding tab **370** define the cable exit angle at between 30° and 60° relative to the (horizontal) circuit card **302**, such as at approximately 45° to immediately lift the cable **310** off of the circuit card **302** and allow tight spacing of the cable assemblies **300** as compared to conventional cable connector modules having the cables oriented generally parallel to the circuit card for termination to the circuit card.

The ground clip **314** is used to mechanically and electrically connect the cable **310** to the circuit card **302**. The mounting tabs **348** mechanically secure the ground clip **314** to the circuit card **302** holding the bottom ground rake **316** along the upper surface **380** of the circuit card **302**. In an exemplary embodiment, the ground clip **314** is used to direct the cable **310** away from the circuit card **302** at a predetermined cable exit angle. For example, the ground clip **314** may hold the cable **310** at a cable exit angle that is non-parallel to the circuit card **302** and non-perpendicular to the circuit card **302**. In various embodiments, the ground clip **314** may hold the cable **310** at a cable exit angle of between 30° and 60°. Optionally, the ground clip **314** may hold the cable **310** at a cable exit angle of approximately 45°. The ground clip **314** directs the cable **310** away from the upper surface **380** at an angle to allow tight spacing of the cable assemblies **300**. For example, the contact pads **384** on the circuit card **302** may be more densely populated (for example, closer spacing) by forcing the cable **310** to exit at an angle from the upper surface **380** of the circuit card **302** as compared to conventional cable connector modules having the cables oriented generally parallel to the circuit card for termination to the circuit card.

In an exemplary embodiment, the ground clip **314** is used to electrically connect to the cable **310**, such as to improve electrical performance of the cable connector module **104**. For example, the ground clip **314** may reduce excess insertion loss and cross talk due to tighter control of electromagnetic fields at the termination area. The ground clip **314** positions the cable **310** to have a short ground return path between the cable **310** and the circuit card **302** for improved electrical characteristics. For example, the ground return path is defined from the cable shield **326** directly into the ground clip **314** through the drain wires **329** and the drain wire tines **344**, and directly from the ground clip **314** to the circuit card **302** through the mounting tabs **348**. The drain wire tines **344** may be soldered to the drain wires **329**. The drain wire tines **344** and the drain wires **329** provide multiple points of contact with the cable shield **326** at different sides of the cable **310** to reduce insertion loss and crosstalk by controlling electromagnetic fields around the end of the cable **310**. The exposed portions **321**, **323** of the signal conductors **320**, **322** have a short distance from the end **325** of the insulator **324** (shown in FIG. 6) to the contact pads **384**. The conductor support **312** tightly controls the impedance in the termination area (between the end **325** of the insulator **324** and the contact pads **384**). The ground clip **314**

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provides shielding in the termination area. For example, the ground clip 314 occupies much of the surrounding space between the end 325 of the insulator 324 and the upper surface 380 of the circuit card 302 to reduce insertion loss and crosstalk by tightly controlling the electromagnetic fields in the termination area.

FIG. 14 is a top perspective view of a portion of the cable connector module 104 in accordance with an exemplary embodiment. FIG. 14 shows the top ground hood 318 coupled to the bottom ground rake 316 and the circuit card 302. In an exemplary embodiment, the top ground hood 318 includes a cable connector 371 configured to be coupled to the cable 310. The cable connector 371 presses the drain wires 329 toward the cable shield 326.

The cable connector 371 extends from the upper grounding tab 370 and wraps at least partially around the end of the cable 310. In an exemplary embodiment, the cable connector 371 includes connecting fingers 373, 375 at opposite sides of the cable connector 371. The connecting fingers 373, 375 may extend from opposite sides of the upper grounding tab 370. The connecting fingers 373, 375 may be provided at or near the distal end of the upper grounding tab 370.

In an exemplary embodiment, the connecting fingers 373, 375 may be folded or crimped around the sides of the cable 310. The connecting fingers 373, 375 are configured to be coupled to the drain wires 329 at the sides of the cable 310. The connecting fingers 373, 375 may directly engage the exposed portions of the drain wires 329 to electrically connect to the drain wires 329. Optionally, the connecting fingers 373, 375 may directly engage the cable shield 326 to electrically connect to the cable shield 326. The connecting fingers 373, 375 are used to pinch the drain wires 329 inward, such as to hold the drain wires 329 against the cable shield 326. In an exemplary embodiment, the connecting fingers 373, 375 are crimped around the end of the cable 310 to press the drain wires 329 inward toward the cable shield 326. In an exemplary embodiment, the connecting finger 373, 375 extend a majority of a perimeter of the cable 310 between the upper grounding tab 370 and the lower grounding tab 342. The connecting fingers 373, 375 may engage the lower grounding tab 342. The connecting fingers 373, 375 may press the lower grounding tab 342 inward toward the cable shield 326.

In an exemplary embodiment, the connecting fingers 373, 375 may be located proximate to the end of the cable shield 326 to connect the drain wires 329 to the cable shield 326 proximate to the end of the cable shield 326. The connecting fingers 373, 375 prevent lift-off or separation of the drain wires 329 from the cable shield 326, such as to create an improved ground return path from the cable shield 326 to the drain wires 329.

In an exemplary embodiment, the cable connector 371 is integral with the top ground hood 318. For example, the cable connector 371 is stamped and formed with the top ground hood 318 from a common metal sheet. However, in alternative embodiments, the cable connector 371 is separate and discrete from the top ground hood 318. For example, the cable connector 371 may be separately stamped and formed. The cable connector 371 may be coupled to the top ground hood 318, such as to the upper grounding tab 370 and/or the lower grounding tab 342. The cable connector 371 may be soldered or welded to the upper grounding tab 370. In alternative embodiments, the cable connector 371 may be crimped onto the upper grounding tab 370. In various embodiments, the cable connector may be a band wrapped entirely circumferentially around the cable 310 and the grounding tabs 370, 342 to compress the drain wires 329 and

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the grounding tabs 370, 342 inward toward the cable shield 326, such as to press the drain wires 329 and the grounding tabs 370, 342 into direct contact with the cable shield 326 to create multiple grounding points with the cable shield 326.

It is to be understood that the above description is intended to be illustrative, and not restrictive. For example, the above-described embodiments (and/or aspects thereof) may be used in combination with each other. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from its scope. Dimensions, types of materials, orientations of the various components, and the number and positions of the various components described herein are intended to define parameters of certain embodiments, and are by no means limiting and are merely exemplary embodiments. Many other embodiments and modifications within the spirit and scope of the claims will be apparent to those of skill in the art upon reviewing the above description. The scope of the invention should, therefore, be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled. In the appended claims, the terms "including" and "in which" are used as the plain-English equivalents of the respective terms "comprising" and "wherein." Moreover, in the following claims, the terms "first," "second," and "third," etc. are used merely as labels, and are not intended to impose numerical requirements on their objects. Further, the limitations of the following claims are not written in means-plus-function format and are not intended to be interpreted based on 35 U.S.C. § 112(f), unless and until such claim limitations expressly use the phrase "means for" followed by a statement of function void of further structure.

What is claimed is:

1. An electronic assembly comprising:

a socket assembly including a cage configured to be mounted to a circuit board, the cage having a cavity, the socket assembly including a socket connector received in the cavity of the cage, the socket connector including a substrate holding interposer contacts, each interposer contact including an upper contact portion and a lower contact portion, the upper contact portion being compressible, the lower contact portion being compressible, the lower contact portion being terminated to a board contact of the circuit board; and

a cable connector module received in the cavity of the cage and coupled to the socket connector, the cable connector module including a housing having a cavity, the housing being coupled to the cage, the cable connector module including a circuit card held by the housing, the circuit card having signal contact pads at an upper surface of the circuit card and mating contact pads at a lower surface of the circuit card, the mating contact pads being coupled to the upper contact portions of corresponding interposer contacts, the cable connector module including cable assemblies coupled to the circuit card, each cable assembly including a cable, the cable having an insulator holding a first signal conductor and a second signal conductor, the cable having a cable shield surrounding the insulator, the first and second signal conductors having exposed portions extending forward of an end of the insulator at an end of the cable electrically connected to corresponding signal contact pads.

2. The cable assembly of claim 1, wherein the cage includes a top opening at the top of the cage, the cable connector module being top loaded into the cavity of the

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cage through the top opening, the mating contact pads being coupled to the upper contact portions from above.

3. The cable assembly of claim 1, wherein the interposer contacts are compressed by the cable connector module.

4. The cable assembly of claim 1, wherein the interposer contacts are dual compression contacts, wherein both the upper contact portions and the lower contact portions are compressed by the cable connector module.

5. The cable assembly of claim 1, further comprising a spring clip coupled to the cage, the spring clip engaging the housing of the cable connector module to hold the cable connector module in the cavity of the cage, the spring clip pressing the cable connector module toward the socket connector.

6. The cable assembly of claim 5, wherein the housing extends between a top and a bottom, the circuit card located at the bottom, the spring clip engaging the top to press downward against the top.

7. The cable assembly of claim 6, wherein the upper wall includes a pocket, the spring clip being received within the pocket.

8. The cable assembly of claim 5, wherein the cage includes a first side wall, a second side wall opposite the first side wall, and an end wall between the first side wall and the second side wall, the cable connector module located between the first side wall and the second side wall, the spring clip being coupled to the first side wall and the second side wall above the cable connector module.

9. The cable assembly of claim 5, wherein the spring clip includes a spring beam extending between a first side and a second side of the spring clip, the first and second sides being coupled to the cage, the spring beam being curved between the first and second sides to engage the cable connector module.

10. The cable assembly of claim 5, wherein the spring clip includes latching features latchably coupled to the cage.

11. The cable assembly of claim 1, wherein the cage includes a latching element, the housing including a latching element interfacing with the latching element of the cage to secure the housing in the cavity of the cage.

12. The cable assembly of claim 1, wherein the cage includes a mounting tab, the mounting tab being coupled to the circuit board to fix the cage relative to the circuit board.

13. The cable assembly of claim 1, wherein the cage includes a window, the cables passing through the window.

14. The cable assembly of claim 1, wherein each cable assembly includes a ground clip coupled to the end of the cable, the ground clip being electrically connected to the cable shield to electrically connect the cable shield to the circuit board, the ground clip supporting the cable such that the cable extends from the circuit board in a cable exit direction angled transverse relative to a mounting surface of the circuit card.

15. The cable assembly of claim 14, wherein the ground clip supports the cable at the cable exit direction of between 300 and 60° relative to the mounting surface of the circuit card.

16. An electronic assembly comprising:

a socket assembly including a cage configured to be mounted to a circuit board, the cage having walls

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extending between a top and a bottom of the cage, the walls defining a cavity, the socket assembly including a socket connector received in the cavity at the bottom of the cage, the socket connector configured to be mounted to the circuit board, the socket connector including a substrate holding interposer contacts, each interposer contact including an upper contact portion and a lower contact portion, the lower contact portion being terminated to a board contact of the circuit board;

a cable connector module received in the cavity of the cage and coupled to the socket connector, the cable connector module including a housing having a cavity, the cable connector module including a circuit card held by the housing, the circuit card having signal contact pads at an upper surface of the circuit card and mating contact pads at a lower surface of the circuit card, the mating contact pads being coupled to the upper contact portions of corresponding interposer contacts, the cable connector module including cable assemblies coupled to the circuit card; and

a spring clip coupled to the walls of the cage, the spring clip engaging the cable connector module to hold the cable connector module in the cavity of the cage, the spring clip pressing the cable connector module into the socket connector to mate the mating contact pads with the upper contact portions of the interposer contacts.

17. The cable assembly of claim 16, wherein the cage includes a top opening at the top of the cage, the cable connector module being top loaded into the cavity of the cage through the top opening, the mating contact pads being coupled to the upper contact portions from above.

18. The cable assembly of claim 16, wherein the interposer contacts are dual compression contacts, wherein both the upper contact portions and the lower contact portions are compressed by the cable connector module.

19. The cable assembly of claim 16, wherein the housing extends between a top and a bottom, the circuit card located at the bottom, the top including an upper wall, the spring clip engaging the upper wall to press downward against the upper wall.

20. The cable assembly of claim 16, wherein the spring clip includes a spring beam extending between a first side and a second side of the spring clip, the first and second sides being coupled to the cage, the spring beam being curved between the first and second sides to engage the cable connector module.

21. The cable assembly of claim 16, wherein each cable assembly including a cable and a ground clip coupled to an end of the cable, the ground clip coupling the cable to the circuit card, the cable having an insulator holding a first signal conductor and a second signal conductor, the cable having a cable shield surrounding the insulator, the first and second signal conductors having exposed portions extending forward of an end of the insulator at an end of the cable soldered to corresponding signal contact pads, the ground clip being electrically connected to the cable shield to electrically connect the cable shield to the circuit card.

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